

Practical Experiments in

BIOLOGY

1



Satya Naidu



Quality ready-to-use
resources



Practical Experiments in

BIOLOGY

1



Satya Naidu



Acknowledgements

I wish to thank Carol Andrews for reviewing the original drafts and Pam Robinson for her suggestions and for providing materials for the experiments.

K. Satya Naidu

© Satya Naidu and Learning Essentials, 2004

Downloadables sold with a licence to print and photocopy

The contents of this book are covered by copyright laws in Australia and in other countries.

The purchase price includes the right for you to print and photocopy these pages for your students without further payment to the publisher. For further information about how schools can use this photocopiable resource, please refer to the 'FAQ's on www.blake.com.au.

Otherwise, you may only download, print, copy, store and communicate the contents of this book without permission when that is allowed by the copyright laws applicable to you.

If you work in an Australian school that is covered by the education statutory licence, you may be allowed to copy, download, print, store and communicate parts of this book without permission from copyright owners. Under the education statutory licence, payments are made on behalf of schools to Copyright Agency, which distributes the payments to writers, artists and publishers. See www.copyright.com.au.

Copyright Agency
Level 12, 66 Goulburn Street
Sydney NSW 2000
Telephone: +612 9394 7600
Facsimile: +612 9394 7601
E-mail: memberservices@copyright.com.au

Publisher: Susan Watson
Editor: Ruth Schultz
Page design: Ruth Schultz
Page formatting and diagrams: Replika Press
Illustrations: Infographics Pty Ltd
Cover design: Gail McManus
Printed in Australia by Imprinta Pty Ltd, Hawthorn

learning
ESSENTIALS

Bringing you the best!

Instant Lessons
An imprint of Learning Essentials
ABN 56 093 980 726
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9548 0193

ISBN 978-1-92561-010-9

INTRODUCTION

Practical Experiments in Biology Book 1 is part of a four-book series that presents experiments from different areas of study that are relevant to senior Biology students. Book 1 focuses on cells, cell processes and factors affecting cell processes.

Teacher’s notes and useful websites for each experiment are provided at the start of the book. These include strategies for enhancing student understanding of the experiment and suggestions for overcoming difficulties. Internet addresses for relevant websites are also provided to encourage students to undertake further research.

The experiments are presented in a self-contained format that Biology students will find easy to follow. Each experiment includes a summary of relevant background knowledge and is structured according to aim, hypothesis, equipment, safety rules, method, results, analysis of results and conclusion. Tables and space for diagrams are provided for students to record their results. The clear, step-by-step structure will encourage the efficient use of experimental time and give guidance to students on how to write scientific reports. Suggested answers and expected results have also been provided where appropriate.

CONTENTS

Teacher's notes and useful websites	4
1 Introduction to light microscope	6
2 Observing cells	10
3 Diffusion and osmosis	14
4 Plasmolysis	18
5 Effect of surface area to volume ratio on cell size	20
6 Factors affecting cell membrane permeability	24
7 Factors affecting catalase activity	28
8 Factors affecting the rate of photosynthesis	32
9 Aerobic and anaerobic respiration	36
10 Suggested answers	39

TEACHER'S NOTES AND USEFUL WEBSITES

Experiment 1: Introduction to light microscope

Suggestions for teachers

- Students should have an introductory lesson on the use of the microscope. Make sure they can identify and describe the function of each part of the microscope.
- Ask students to always start with low-power objective before using the high-power objective.
- Remind students that a nanometre (nm) is 10^9 of a metre, a micrometre is 10^6 of a metre and the resolution of a light microscope is about 200–250 nm.

Websites

<http://www.stanford.edu/group/Urchin/size.htm>
<http://faculty.clintoncc.suny.edu/faculty/Michael.Gregory/files/Bio%20101/Bio%20101%20Laboratory/Microscopy/Microscopy%20lab.htm>
http://www.umanitoba.ca/Biology/lab1/biolab1_7a.html
<http://www.enchantedlearning.com/label/>

Experiment 2: Observing cells

Suggestions for teachers

- Students should be familiar with using the microscope before attempting this experiment.
- Students should have good background knowledge about prokaryotic and eukaryotic cells.
- Students' diagrams do not have to be detailed, but should show the major parts and appropriate labels. It is important for students to learn to draw what they actually see, as opposed to replicating a diagram they may have seen in a textbook.
- Show a video on cells and use an overhead projector or a camera attached to the microscopes to enlarge the cells.

Websites

http://www.umanitoba.ca/Biology/lab1/biolab1_6.html#Prepared
<http://www.uleth.ca/bio/bio1010/bio1010.htm>
<http://www.microscopy-uk.org.uk/mag/indexmag.html>
<http://www.microscopy-uk.org.uk/mag/articles/blood1.html>
<http://www.sidwell.edu/us/science/vlb7/tissuelab/results/blall.html>
http://www.umanitoba.ca/Biology/lab1/biolab1_7a.html

Experiment 3: Diffusion and osmosis

Suggestions for teachers

- Before beginning this activity, the students should know how to use an electronic balance and draw line graphs.
- Students should have background knowledge of diffusion/osmosis and the differences between hypotonic, isotonic and hypertonic solutions.
- Cellulose tubing should be soaked in tap water in advance so that it opens easily.

- Tes-tape is available from pharmacies; it is used to test the urine of diabetics for the presence of glucose.

Websites

<http://faculty.clintoncc.suny.edu/faculty/Michael.Gregory/files/Bio%20101/Bio%20101%20Laboratory/Cells/cells.htm>
<http://users.rcn.com/jkimball.ma.ultranet/BiologyPages/D/Diffusion.html>
www.personal.psu.edu/faculty/t/r/trp2/diffusion.html
www.edvotek.com/pdfs/281.pdf

Experiment 4: Plasmolysis

Suggestions for teachers

- Before doing this experiment, students should be aware of the process of plasmolysis and deplasmolysis.
- Students should be familiar with preparing wet mounts and focusing the microscope.
- Onion cells contain purple reflecting pigments within the plasma membrane. The shrinking of the membrane can be seen easily.
- Students should draw and label only what they see through the microscope.

Websites

http://discoveryscience.net/Class_Handouts/Osmosis_lab.pdf
http://www.personal.psu.edu/faculty/w/x/wxm15/Online/Cyto_unit/membrane_transport/aca_lec02_.htm
eve.kean.edu/~breid/plantlab2/plant_2.html
http://eve.kean.edu/~breid/plantlab2/plant_2.html

Experiment 5: Effect of surface area to volume ratio on cell size

Suggestions for teachers

- Discuss the importance of models in understanding the concept of diffusion. A model is just a representation of reality that simplifies things and represents the most important aspects.
- Emphasise that changes in the surface area to volume ratio have important implications in constraining an organism's size.
- Remind students that the acid diffuses in each cube about 0.5 cm from each side.

Websites

http://www.tiem.utk.edu/~gross/bioed/bealsmodules/area_volume.html
<http://www.geocities.com/CapeCanaveral/Hall/1410/lab-B-24.html>
http://www.accessexcellence.org/AE/AEC/AEF/1996/deaver_cell.html
<http://www.mrshelley.com/Honors%20Biology/surface%20area%20to%20volume%20of%20cell%20lab.htm>

Experiment 6: Factors affecting the cell membrane permeability

Suggestions for teachers

- Students should have good background knowledge of the structure of the cell membrane and the factors that affect the cell membrane.
- During the pre-experiment discussion, guide the students to point out that the independent variable is the intactness of the cell membrane, as measured by its permeability to the red pigment.
- The relative colour intensity of the solution will be a valid indication of the permeability of the cell membrane. The intensity of the red colour of the solutions can be compared by viewing the solutions against a white background.
- If a colorimeter is available, it will give good quantitative results using a green filter.
- Remind students that detergent and alcohol affect the fatty acids in the cell membrane and pH changes and high temperature denature protein molecules in the membrane.

Websites

<http://biology.arizona.edu/sciconn/lessons/mccandless/tchrinfo.html>
http://mtsu32.mtsu.edu:11289/cyto2_web/sld007.htm
http://academic.brooklyn.cuny.edu/biology/bio4fv/page/pm_mos.htm
http://www.ivy-rose.co.uk/Topics/Cell_Structure.htm
http://www.tulane.edu/~wiser/methods/handouts/class/10_memb.pdf

Experiment 7: Factors affecting catalase activity

Suggestions for teachers

- Before the experiment, students should practise using the electronic balance.
- Students should have background knowledge of enzymes and factors affecting enzyme activity.
- Use fresh liver, fresh chicken and fresh vegetables for testing the presence of catalase.
- Hydrogen peroxide must be made on the day of the experiment. Do not store it in the fridge, as cool temperatures slow down the rate of reaction.

Websites

<http://users.rcn.com/jkimball.ma.ultranet/BiologyPages/E/Enzymes.html>
<http://www.geocities.com/CapeCanaveral/Hall/1410/lab-B-01.html>
<http://www.bjbarton.com/lessons/Catalase.doc>
<http://faculty.clintoncc.suny.edu/faculty/Michael.Gregory/files/Bio%20101/Bio%20101%20Lectures/Energy/energy.htm#Enzymes>

Experiment 8: Factors affecting the rate of photosynthesis

Suggestions for teachers

- Students should have a thorough understanding of the process of photosynthesis and be familiar with factors that affect the rate of photosynthesis.

- Plants must be destarched for a day in a dark cupboard before the experiment. The destarched plants can then be placed in light for 24 hours before doing the starch test.
- A microscope lamp or a 40-watt lamp can be used to test the effect of light on photosynthesis. But control other sources of light to avoid experimental errors. Darken the room as much as possible.
- Care must be taken to make sure that the heat of the lamp does not affect the plant. Take regular temperature measurements of the water in the beaker, which acts as the heat shield. Set up the apparatus so that it is airtight by using petroleum jelly.

Websites

<http://www.teachnet.ie/foneill/exper.htm-animated>
<http://www.geocities.com/CapeCanaveral/Hall/2385/rate.htm>
<http://www.shcc.edu.hk/~is/phs2.htm>
<http://web.ukonline.co.uk/webwise/spinneret/plants/psfac2.htm>
<http://www.learn.co.uk/default.asp?WCI=Unit&WCU=3539>
<http://www.google.com.au/search?hl=en&ie=UTF-8&oe=UTF-8&q=diagram+audus+apparatus&meta=>

Experiment 9: Aerobic and anaerobic respiration

Suggestions for teachers

- Students should have good background knowledge of cellular respiration, glycolysis and fermentation.
- The experimental set up should be airtight to prevent the entry of air. Thermos flasks can be used instead of conical flasks in part B.
- The glucose solution should be boiled and cooled to remove oxygen.
- Activate the yeast by dissolving it in warm water and the adding it to the glucose solution. The foam or froth that is produced and the 'yeasty' aroma are signs that the yeast is actively respiring.
- Wash the pea seeds in bleach to remove bacterial and fungal spores. Peas could be germinated ahead of time to reduce the time of observation.

Websites

<http://www.geocities.com/hkbiology/respirationexperiment.htm>
<http://www.learn.co.uk/default.asp?WCI=Unit&WCU=2352>
 (includes animation)
<http://web.ukonline.co.uk/webwise/spinneret/humans/respro.htm>
<http://www.ns.purchase.edu/biology/bio1550lab/aerobic.htm>
http://www.mrothery.co.uk/bio_web_prac/practicals/2anaerobic%20resp%20student.doc

INTRODUCTION TO LIGHT MICROSCOPE



Timing: 2 × 45 minute lessons

Background

Microscopes have enhanced our knowledge in basic biology, biomedical research, medical diagnostics and material science. The most widely used microscopes are **optical microscopes**. These microscopes use visible light to create a magnified image of an object that is too small to be seen with the unaided eye.

The **compound light microscope** uses two lenses, an objective lens and an ocular lens. The two lenses are mounted at opposite ends of a closed tube to provide greater **magnification**.

Magnification is the extent to which the image of a specimen is enlarged when being viewed under the microscope. The total magnification of a light microscope can be more than 2000 times.

The **resolving power** of a microscope is the ability to see two objects that are close together as separate objects; that is, the ability to observe details. Light microscope usually has a limited resolution of between 200–250 nm. This means that any two points closer than this will appear to be one point.

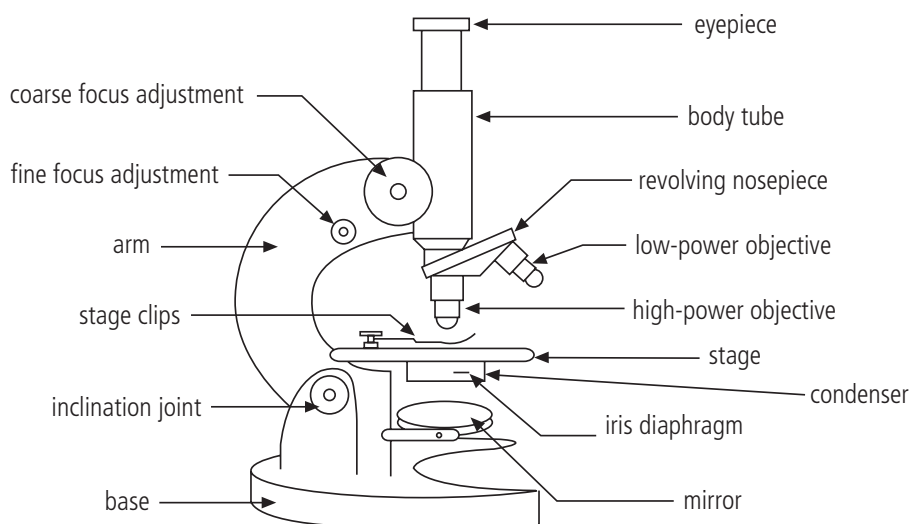
The area of the slide that you see when you look through a microscope is called the **field of view**. If you know the width of the field of view, you can estimate the size of the things you see in the field of view. When switching from low power to high power, the area in the field of view gets smaller and darker. As the magnification increases the field of view decreases.

Aim

To become familiar with handling and using a light microscope and to measure the size of a microscopic organism (paramecium).

Equipment

- newspaper
- coloured photograph from a magazine
- scissors
- cavity slides
- cover slips
- compound light microscope
- microscope lamp
- dissecting needle
- forceps
- 10× and 40× objectives
- 10× eyepiece
- transparent mm ruler
- lens tissue
- toothpicks
- camel hair brush
- culture of paramecium
- ProtoSlo
- cotton fibres
- pipette
- water in a beaker
- paper towel



SAFETY ISSUES

- ⚠ When carrying a microscope, always use one hand to hold the arm and your other hand to support the base.
- ⚠ Never use cloth or paper products to clean the lenses. They will scratch the coating and decrease the resolving power of the lens. Only use lens paper.
- ⚠ Raise the objectives from the stage before inserting a slide. This will prevent the objective lens from being scratched by the slide.

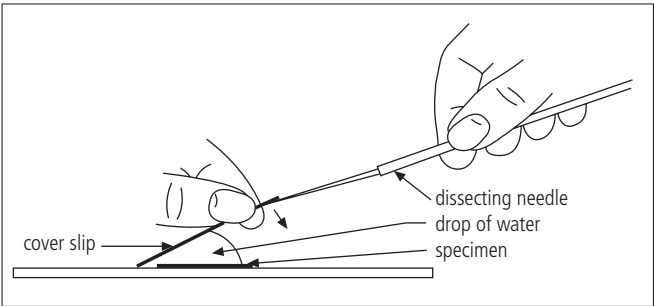
- ⚠ Always focus first using the coarse-focus adjustment with the low-power objective lens in position.
- ⚠ Only use the fine-focus adjustment when the high-power objective lens is in position.
- ⚠ Do not let ProtoSlo get on the objective lenses or the stage. It is very sticky and makes the lenses unusable. If it gets on the stage, it makes the slides stick to the stage.
- ⚠ When storing a microscope, have the low-power objective in position and cover the microscope with a dust cover or place it in its box.

Method

- 1 Place the microscope on your desk. Clean the objectives and eyepieces with tissue paper or a fine camel hairbrush.
- 2 Switch on the microscope lamp and shine the light on to the mirror.
- 3 Use the coarse-focus adjustment to lower the low-power objective (10×) until the lens is 1 cm above the stage.
- 4 Look through the eyepiece and adjust the diaphragm until the view is as bright as possible.

Making a wet mount

- 5 Cut the letter 'e' from a newspaper and use forceps to place it on a slide. Use a pipette to add one drop of water. Place the cover slip at a 45° angle and use a dissecting needle to lower it slowly on the slide so that no air bubbles are trapped.
- 6 Place the slide in the centre of the stage and focus using the low-power objective. Compare the appearance of the letter 'e' you see through the eyepiece with the printed letter 'e'. Draw a diagram.
- 7 While viewing through the eyepiece, move the slide slowly to the left and then right. Observe how the image moves.
- 8 Move the slide slowly up and down. Observe how the image moves.
- 9 Using the coarse-focus adjustment, move the body tube up and rotate the revolving nosepiece so that the high-power objective clicks into place. Make sure the objective does not touch the cover slip.
- 10 Adjust the iris diaphragm to give a brighter light and use the fine-focus adjustment to refocus. Draw a diagram of the letter 'e' viewed through the high-power objective.



Calculation of total magnification and field of view

To calculate the total magnification of an image, multiply the magnification of the eyepiece by the magnification of the objective. Fill in the total magnification in the table below.

Objective magnification	Eyepiece magnification	Total magnification
Low power 10×	Low power 5×	
Low power 10×	High power 10×	
High power 40×	High power 10×	

Field of view

- 11 To determine the field of view, place a transparent plastic mm ruler on the stage. Use the coarse-focus adjustment and 10× objective to focus on the millimetre marks on the ruler. Position the ruler so that a millimetre marking is at the left edge of the field of view.
- 12 Measure the diameter within the field of view by counting the number of millimetres. Remember that 1 mm is 1000 microns; so 1.5 mm equals 1500 microns.

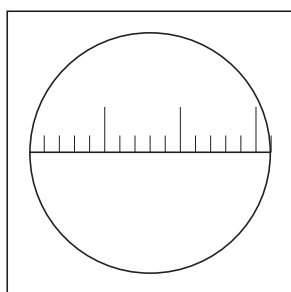
- 13 Remove the ruler. Place a drop of paramecium culture on a cavity slide and add a cover slip. Close down the aperture of the iris diaphragm and use minimal light as the paramecium is nearly transparent. Examine the paramecium culture with low-power and high-power objectives.
- 14 Place a drop of paramecium culture on another cavity slide and add half a drop of ProtoSlo (a very viscous methylcellulose solution). Use a toothpick to gently mix the two solutions together. Add a few cotton fibres and a cover slip. The ProtoSlo and cotton fibres will immobilise the paramecia. As water evaporates from the edge of the cover slip, the paramecia will be trapped under the cover slip and will be easier to observe. Count the number of paramecia you can see across the diameter. Draw a diagram under Results.

Note: Do not let ProtoSlo get on the objective lenses or the stage. ProtoSlo becomes very sticky and makes the lenses unusable. When on the stage, ProtoSlo makes the slides stick so they can't be moved.

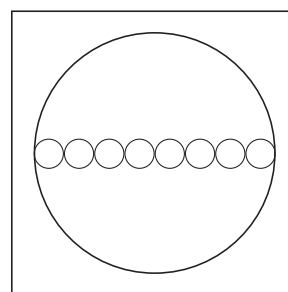
- 15 Calculate the size of one paramecium. For example, if the diameter of the field of view is 1.6 mm or 1600 microns wide, a rough estimate can be made of the size of paramecium in view. If 8 paramecia fit across the diameter, the size of the organism can be calculated as shown:

$$\frac{\text{Diameter of field of view of low power}}{\text{Number of paramecia that fit across the diameter}} = \frac{1600 \text{ microns}}{8 \text{ paramecia}} = \frac{200 \text{ microns}}{1 \text{ paramecium}}$$

The approximate size of one paramecium is 200 microns.



Field of view diameter = 1.6 mm = 1600 microns



Number of paramecia across the diameter = 8

Size of a paramecium = $1600 \div 8 = 200$ microns

- 16 Leave the coarse-focus adjustment in position and shift to 40× objective. Take care that it does not come in contact with the suspension or the cover slip. Use the fine-focus adjustment with the 40× objective. Adjust the diaphragm to control the amount of light.
- 17 Calculate the field of view of the high-power objective. Then find the approximate size of one paramecium. This can be calculated as shown:

The high-power objective (400×) has 0.25 the field of view of the low-power objective (100×).
The diameter of the high power objective is:

$$\begin{aligned} \text{Low-power field of view} \div 4 &= \text{high-power field of view} \\ 1600 \div 4 &= 400 \text{ microns} \end{aligned}$$

Therefore the high-power field of view is 400 microns in diameter. If 4 paramecia fit across the diameter, the size of one paramecium is: $400 \div 4 = 100$ microns

- 18 Fill in the table at Results 3.

Resolving power

- 19 Prepare a wet mount of a small piece of a coloured photograph from a magazine. Examine the mount under low power.
- 20 Compare the detail you can see in the image to the detail you can see in the photograph with your naked eye. This gives you the resolving power of the microscope (the ability of a microscope to clearly separate the small details present in an object as distinct images). Light microscope usually has a limited resolution between 200–250 nm.
- 21 Always return to the low-power objective before you pack up the microscope. Wipe up any spilled water with a cloth or paper towel dampened in tap water. Cover the microscope with a dust cover.
- 22 Clean and dry the slides and cover the slips with a paper towel before placing them in their boxes.

Results

- 1 Draw diagrams of letter 'e' as viewed under low-power objective and high-power objective.
- 2 Draw a diagram to show the number of paramecia seen under low-power objective. Indicate the size of one paramecium in microns.
- 3 Complete the table.

Eyepiece	Objective	Magnification (eyepiece × objective)	Field of view (in microns)	No. of paramecia across field of view	Size of a paramecium (in microns)
10×	10×				
10×	40×				

Analysis of results

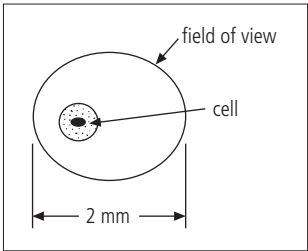
- 1 Copy the table. Then describe the functions of each part of the microscope.

Part of microscope	Function	Part of microscope	Function
Eyepiece		Mirror	
Body tube		Coarse-focus adjustment	
Revolving nosepiece		Fine-focus adjustment	
Low-power objective		Arm	
High-power objective		Stage clips	
Stage		Inclination joint	
Diaphragm		Base	

- 2 What happened to the image when the slide was moved:
a to the right. b to the left. c up. d down.

Conclusion

- 1 Define the following terms.
a magnification b field of view c resolving power
- 2 How did letter 'e' appear under low-power and high-power objectives?
- 3 The picture shows the number of cells seen under low-power field of view.
a What is the approximate size of the cell in micrometres?
b What would be the high-power field of view?
c How many cells would fit in the high-power field of view?



Timing: 4 × 45 minute lessons**Background**

According to cell theory, cells are the smallest functional units of life. All living organisms are composed of cells and cell products. Two basic types of cells are **prokaryotic** cells and **eukaryotic** cells. Prokaryotic cells have a more primitive (pro) nucleus (karyo) without a membrane. Bacteria and blue-green algae are prokaryotic cells. Eukaryotic cells have true (eu) nuclei (karyo) enclosed by a nuclear membrane. Plant and animal cells are eukaryotic cells.

Plant cells have **cell walls** and large **vacuoles**. Only photosynthetic cells have **chloroplasts**. Potato and banana cells store starch in **leucoplasts**. Coloured plant cells have **chromoplasts**. The movement of cytoplasm around the vacuole of a plant cell is known as **cytoplasmic streaming**. This enables cell products to be carried to different parts of the cell.

Animal cells do not have any chloroplasts or cell walls. Some animal cells do not have vacuoles; if vacuoles are present they are smaller in size. Human red blood cells do not have nuclei. This means that they have more surface area for carrying oxygen and they bend easily when passing through fine capillaries. The red blood cells of a frog have nuclei when they mature and divide. The white blood cells of frogs are similar to human in both form and function.

Staining cells enables us to view **cell organelles** more clearly. However, the cells are killed during this process because stains are toxic to cells.

Aim

- To stain and make wet mounts of prokaryotic and eukaryotic cells and note the effects of staining.
- To observe the similarities and differences between prokaryotic and eukaryotic cells.

Equipment

- | | | |
|---------------------------------------|--|-----------------------|
| • light microscope | • cover slips | • camel hair brush |
| • microscope lamp | • onion | • cold water |
| • oil immersion objective | • elodea plant | • warm water |
| • immersion oil | • potato | • 5% sucrose solution |
| • methylene blue stain | • red capsicum | • yogurt |
| • iodine solution (I ₂ KI) | • scalpel | • Bunsen burner |
| • Janus Green B stain | • prepared slides of frog blood and human blood smears | • paper towel |
| • crystal violet stain | | • spray bottle |
| • toothpicks | | • pipettes |
| • microscope slides | • tissue paper | |

SAFETY RULES

- ⚠ Only use a lens tissue or a camel hairbrush to clean the slides and cover slips.
- ⚠ Keep the sample specimens thin (a few cell layers at most) and do not let them dry.
- ⚠ When viewing through high-power objective, only use the fine focus adjustment.
- ⚠ After the experiment, wash the slides and cover slips and dry them before storing. Cover the microscope with a dust cover before returning it to its box.
- ⚠ Be careful using oil immersion objective when viewing lactobacillus. Clean the oil with paper towel.

Method**Part A – Preparing and staining prokaryotic cells****Lactobacillus from yogurt**

- 1 Use a toothpick to place a tiny amount of yogurt on a microscope slide.
- 2 Add one drop of water and mix it in with the yogurt. Allow the bacterial smear to air dry.
- 3 After the smear has dried, hold the slide by one edge and pass it slowly through a Bunsen burner flame.
- 4 Add a few drops of **crystal violet** over the smear and allow the stain to work for 30 seconds.
- 5 Remove excess crystal violet by washing the slide with cold water from the spray bottle. Gently blot the stain dry with a paper towel.

- 6 Place a drop of **immersion oil** on the slide and then view it with an oil immersion objective (100×). Make sure the objective makes contact with the immersion oil. Be careful not to break the slide as you view it.
- 7 Adjust the illumination by closing or opening the iris diaphragm and focus using the fine focus adjustment.
- 8 You should be able to see a number of rod-shaped bacteria. Draw and label bacterial cells as they appear under oil immersion objective.
- 9 Rotate the objective to the side and use lens paper to clean any remaining oil.

Part B – Preparing and staining eukaryotic plant cells

Onion cells

- 1 Cut an onion bulb into quarters and remove one of the fleshy leaves. Break the leaf backwards to produce a ragged section of epidermis.
- 2 Place a drop of water on a slide. Remove a small piece of the epidermis and spread it evenly in the water. Lower a cover slip gently onto the slide to prevent trapping air bubbles.
- 3 Examine the slide with your microscope. If the slide begins to dry, add more water to the edge of the cover slip with an eye dropper.
- 4 Add 1 or 2 drops of **methylene blue** to the edge of the cover slip. Draw the stain under the cover slip by touching lens paper to the opposite side of the slip.
- 5 Draw and label the stained onion cells as viewed under high-power objective.

Observing mitochondria in onion cells

- 6 Prepare a second onion slide, but use one drop of **5% sucrose solution** instead of water.
- 7 Add 3 drops of **Janus Green B** stain to the edge of the cover slip. Draw the stain under the cover slip by touching lens paper to the opposite side of the slip.
- 8 View the slide at 400×. The mitochondria should appear as small rods and spheres at the periphery of the cell. They will be stained blue in colour. If you don't see any mitochondria, add 3 more drops of Janus Green B stain to the other edge of the cover slip.

Elodea cells

- 9 Take a leaf from an Elodea plant. Place it on a slide so that the upper part of the leaf is in position. Add one drop of water and put the cover slip on carefully. Observe the cells through low-power and high-power objectives.
- 10 Add one drop of warm water to the slide. Observe if the cytoplasm is moving around the vacuole. This process is known as **cytoplasmic streaming**.
- 11 Add one drop of **iodine** to the elodea leaf and remove the excess with a piece of paper towel. Observe the stained elodea cells through low-power and high-power objectives.
- 12 Draw and label stained elodea cells as viewed under high-power objective. Note what happens to cytoplasmic streaming after adding the stain.

Potato cells

- 13 Cut a slice of potato. Place one drop of water in the centre of a slide. Use a toothpick to scrape a thin layer from the cut surface of the potato and spread it carefully on the slide.
- 14 Add a cover slip and observe the slide under low-power and high-power objectives.
- 15 Irrigate your slide with **iodine solution** and observe the stained **leucoplasts** which appear purple in colour.
- 16 Draw and label the stained potato cells as viewed under high-power objective.

Red capsicum cells

- 17 Cut a very thin slice of red capsicum and place it on a slide.
- 18 Add one drop of water and then the cover slip. Observe the **chromoplasts** which appear orange.
- 19 Draw and label red capsicum cells as viewed under high-power objective.

Part C – Preparing and staining eukaryotic animal cells

Human cheek epithelial cells

- 1 Gently scrape the inside of your cheek with the broad end of a toothpick. Stir the scrapings into a drop of water on a slide.
- 2 Add a cover slip and examine the slide. Close the diaphragm to reduce light as needed.
- 3 Irrigate the cheek cells with **methylene blue stain** and observe the stained cheek cells under low-power and high-power objectives.
- 4 Draw and label stained cheek cells as viewed under high-power objective.

Human and frog blood cells

- 5 Observe a prepared slide of human blood cells through low-power and high-power objectives. These cells have been stained with a dye similar to methylene blue to make red blood cells appear pink or red and white blood cells appear dark blue or purple.
- 6 If white blood cells are visible, note the differences between red blood cells and white blood cells. Draw diagrams of red blood cells and white blood cells as viewed under high-power objective.
- 7 Observe a prepared slide of frog blood cells through low-power and high-power objectives. Draw and label the parts of blood cells as viewed under high-power objective.
- 8 Note the differences between the following cells: prokaryotic and eukaryotic cells; human and frog blood cells; stained and unstained cells; plant and animal cells.

Results

- 1 On a separate sheet of paper, draw and label stained cells as viewed under high-power. Draw diagrams of Lactobacillus stained 1000x; Onion cells stained 400x; Elodea cells stained 400x; Potato cells stained 400x; Red capsicum unstained 400x; Cheek cells stained 400x; Human blood cells stained 400x; Frog blood cells stained 400x.
- 2 Use ticks and crosses to fill in Table 1.

Table 1 Organelles seen in prokaryotic cells and eukaryotic cells

Cell organelles	Prokaryotic cells	Eukaryotic plant cells				Eukaryotic animal cells		
	Lactobacillus	Onion cells	Elodea cells	Potato cells	Red capsicum cells	Cheek cells	Human red blood cells	Frog red blood cells
Cell wall								
Cell membrane								
Cytoplasm								
Nucleus								
Mitochondria								
Chloroplasts								
Leucoplasts								
Chromoplasts								
Vacuole								

- 3 Use ticks and crosses to fill in Table 2.

Table 2 Differences between plant and animal cells

Cell organelle	Plant cell	Animal cell
Cell wall		
Chloroplasts		
Vacuole		

Analysis of results

Prokaryotic cells – lactobacillus

- 1 Describe the structure of lactobacillus.
- 2 Which organelles did you see in stained bacterial cells?
- 3 How are prokaryotic cells different from eukaryotic cells?

Eukaryotic cells

Onion cells

- 4 Describe the shape of an onion epidermal cell.
- 5 Which organelles did you see in onion cells stained with potassium iodide and Janus Green B?

Elodea cells

- 6 Which organelles did you see in elodea cells stained with potassium iodide?
- 7 Did you observe cytoplasmic streaming in unstained elodea cells? What is the purpose of this process?
- 8 What happened when iodine was added to elodea cells?

Potato cells

- 9 Which plastids do potato cells contain? What do they store?
- 10 What happened when iodine was added to potato cells?

Red capsicum cells

- 11 What type of plastids did you see in red capsicum cells?
- 12 What is the function of these plastids?

Cheek epithelial cells

- 13 Describe the shape of cheek cells. What is their function?
- 14 Which cell organelles did you see in stained cheek cells?

Human and frog red blood cells

- 15 What differences could you see between the red blood cells of a frog and a human?
- 16 How were human white blood cells different from human red blood cells?
- 17 From your results, how could you distinguish between a plant cell and an animal cell?

Effect of staining

- 18 Describe the effect of staining on the following cells: Crystal violet to lactobacillus; Potassium iodide added to onion, elodea and potato cells; Janus green B stain added to onion cells; Methylene blue added to cheek and blood cells.
- 19 Describe the functions of these cell organelles: Cell wall, Cell membrane, Cytoplasm, Nucleus, Vacuole, Chloroplast, Chromoplast, Leucoplast, Mitochondria. Use your notes and reference books to help you

Conclusion

- 1 Write a paragraph explaining why cells have different shapes and contain different cell organelles.
- 2 If you were given a slide containing living cells of an unknown organism, how would you identify whether they are plant cells or animal cells?
- 3 What differences did you notice between prokaryotic cells and eukaryotic cells?

Background

Timing: 2 × 45 minute lessons

Diffusion is the passage of molecules from a region of high concentration through a semi-permeable membrane to a region of low concentration. **Osmosis** is the passage of water from a region of high water concentration to a region of low water concentration. **Semi-permeable membranes** are very thin layers of material which allow smaller molecules such as water, potassium iodide and glucose to pass through them, but prevent larger starch molecules from passing through. In a **hypertonic solution**, the solute concentration is higher than that in cells. In a **hypotonic solution**, the solute concentration is below that in cells. In an **isotonic solution**, the solute concentration is equal inside and outside the cell.

In this experiment, dialysis tubing is used to demonstrate the action of a differently permeable membrane. When plant cells are placed in **hypertonic solution**, they lose water and become **flaccid**. When plant cells are placed in **hypotonic solution**, they gain water and become turgid. In an **isotonic solution**, cells are **unchanged** because the amount of water leaving the cell is equal to the amount of water entering the cell.

Aim

- To investigate how starch, glucose molecules and water molecules diffuse through a dialysis tubing which is acting as a differently permeable membrane.
- To observe the process of osmosis when potato cells are placed in hypertonic, hypotonic and isotonic salt solutions.

Equipment

- dialysis tubing
- scissors
- soluble starch solution
- iodine solution
- glucose solution
- 2 thistle funnels
- rubber band
- 2 retort stands with clamps
- 2 gas jars
- tes-tape
- iodine solution (I_2KI)
- ruler
- 2 glass jars
- waterproof marking pen

SAFETY RULES

- ⚠ Be careful while using the cork borer and scalpel.
- ⚠ Do not discard potato cylinders, dialysis tubing or solutions in the sink.
- ⚠ Wash your hands after using iodine solution.

Part A – Diffusion and osmosis through a differentially permeable membrane

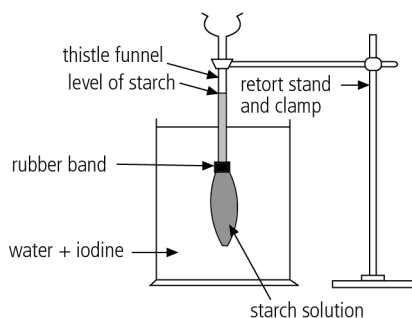


Figure 1: Jar A – Dialysis tubing with starch

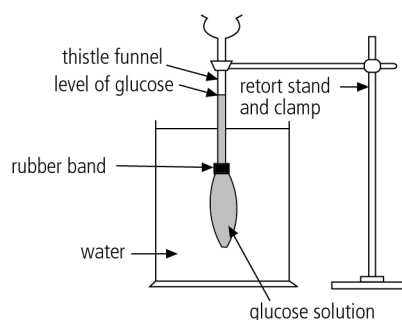


Figure 2: Jar B – Dialysis tubing with glucose

Method

- 1 Moisten the dialysis tubing with warm water and then open it by rubbing it between your thumb and forefinger.
- 2 Seal one end of the tubing by tying a knot 1 cm from the end. Take care not to damage the tubing.
- 3 Slide the thistle funnel into the open end of the dialysis tubing. Fix the bag tightly to the thistle funnel by winding the rubber band around it securely.
- 4 Pour starch solution into the thistle funnel until the dialysis tube is filled and the starch solution rises 3 cm up the stem of the funnel. Make sure there are no air bubbles.

- 5 Rinse the outside of the thistle funnel and dialysis tubing with cold water to remove starch.
- 6 Suspend the bag in the gas jar and fix the apparatus in place with a retort stand and clamp as shown in figure 1.
- 7 Add water to the gas jar until the level is 1 to 2 cm above the level of the bag.
- 8 Pour enough **iodine solution** into the water to give it a yellowish colour.
- 9 Label the jar 'A'. Use a waterproof marking pen to mark the level of the water in the gas jar and the level of the starch solution in the thistle funnel stem.
- 10 Leave the apparatus undisturbed for one hour. Note colour changes in the gas jar and dialysis tubing.
- 11 Record any changes in the water level in the gas jar and the stem of the thistle funnel.
- 12 Set up a second apparatus in the same way, but add glucose instead of starch in the dialysis tubing. Do not add iodine to the water in the gas jar.
- 13 Label the second apparatus 'B' and mark the level of glucose solution in the thistle funnel and the level of water in the gas jar.
- 14 Dip a piece of **tes-tape** into the water in the jar and record colour change. Leave the apparatus undisturbed for one hour.
- 15 Observe the level of glucose solution in the tube and the water level in jar 'B'. Measure and record any changes. Dip a piece of tes-tape into the water in the jar and record colour change. **Note:** Remember that iodine turns starch blue-black and tes-tape turns green in the presence of glucose.

Results

Table 1 Starch test

Experimental set up	Colour of water in the jar before and after diffusion	Colour of starch in the tubing before and after diffusion	Level of water in gas jar before and after diffusion	Level of water in the thistle funnel before and after diffusion	Interpretation
Jar A					

Table 2 Glucose test

Experimental set up	Colour of tes-tape in the gas jar before and after diffusion	Colour of tes-tape in the gas jar before and after diffusion	Level of water in gas jar before and after diffusion	Level of water in the thistle funnel before and after diffusion	Interpretation
Jar B					

Analysis of results

- 1 What caused the colour change in the starch solution when the dialysis tubing was left in the beaker containing iodine solution?
- 2 Did starch molecules diffuse through the dialysis tubing in set up A? Explain.
- 3 Did glucose molecules diffuse through the cellophane tubing in set up B? What was the evidence that it was diffusing?
- 4 Was there evidence of water molecules moving across the dialysis tubing in gas jars A and B?



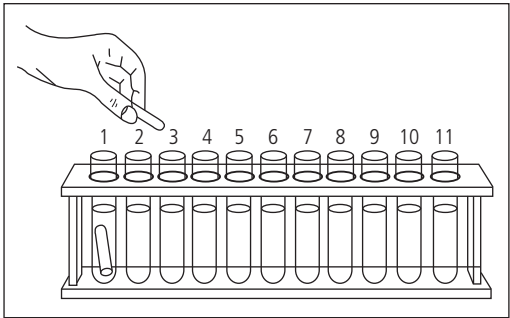
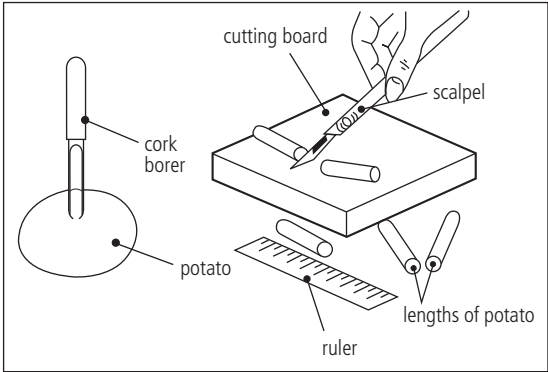
Part B – Osmosis in potato cells placed in salt solutions of different concentrations

- Equipment**
- Salt solutions of different concentrations (10% to 100%)
 - distilled water
 - cork borer (d = 1 cm)
 - scalpel
 - one large peeled potato
 - 11 test tubes
 - test tube rack
 - 10 mL measuring cylinder
 - ruler
 - electronic balance
 - dissecting needle
 - paper towelling
 - marking pen.

Method

- 1 Use a cork borer and a scalpel to cut 11 potato cylinders measuring 2 cm in length. Make sure there is no potato skin on the cylinders.
- 2 Use an electronic balance to weigh each potato cylinder. The weight and length of all potato cylinders should be the same. Record the length and weight of each cylinder in Table 1.
- 3 Label eleven test tubes as shown in the table below.

Test tube 1	Test tube 2	Test tube 3	Test tube 4	Test tube 5	Test tube 6	Test tube 7	Test tube 8	Test tube 9	Test tube 10	Test tube 11
Distilled water	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Salt solutions										



- 4 Place 1 potato cylinder in each of the test tubes numbered 1–11.
- 5 Pour 10 mL of distilled water into test tube 1. Pour 10 mL of salt solutions of different concentrations (as shown in the table above) in test tubes 2–11.
- 6 Leave the potato cylinders in the salt solutions for 30 minutes.
- 7 After 30 minutes, use a dissecting needle to remove the potato cylinders from the test tubes. Place them on paper towel and pat them dry.
- 8 Measure the length and weight of each potato cylinder and record your measurements in Table 3.
- 9 Observe any changes in the appearance of the potato cylinders. Record in Table 3 whether they are turgid or flaccid.
- 10 Calculate the change in mass as a percentage of the original mass by using the formula given below.

$$\% \text{ change in mass} = \text{mass change (+ive or -ve)} \times 100 \div \text{initial mass}$$

(If it is a gain on mass, record it as a positive percentage. If it is a loss on mass, record it as a negative percentage.)

- 11 Collate the class results and calculate the average % change in mass for each solution.
- 12 Discard the potato cylinders in a rubbish bin and clean the test tubes.

Results

Table 3 Changes in length and mass

Test tube no.	Solution used	Changes in length and mass of potato cylinders						
		Length before (a) mm	Length after (b) mm	Change in length (b – a) mm	Mass before (c) g	Mass after (d) g	Percentage change in mass $(c - d) \div c \times 100$	Interpretation
1	Distilled water							
2	10%							
3	20%							
4	30%							
5	40%							
6	50%							
7	60%							
8	70%							
9	80%							
10	90%							
11	100%							

Graphing

Using the class data, draw a line graph by plotting the change in mass of potato cylinders against the concentration of sodium chloride solution.

Analysis of results

Use the class results to answer these questions.

- 1 Which test tube was used as a control with the potato cylinders?
- 2 What were the dependent and independent variables?
- 3 Which salt concentration caused the greatest changes in mass and length? Explain.
- 4 Which salt solution caused the smallest changes? Explain.
- 5 If the concentration of potato cell sap was 15%, what would you expect to see if it is placed in 15% salt solution?
- 6 List three possible errors that could have occurred in this experiment. Include a valid explanation for why each one might have occurred.

Conclusion

- 1 What is the difference between diffusion and osmosis?
- 2 Which molecules diffused through the cellophane tubing in part A?
- 3 What happened in the case of potato cylinders placed in different concentrations of salt solutions?

Background

Timing: 2 × 45 minute lessons

If a plant cell is immersed in a **hypertonic** solution, water will leave the cell. This process is called **plasmolysis**. The loss of water causes the cell to lose the pressure exerted by the fluid in the plant cell's vacuole (called turgor pressure) and the plant cell becomes flaccid. During plasmolysis, the cellular membrane pulls away from the cell wall and shrinks.

When plant cells are placed in **hypotonic** solution, water enters the cell and the cells become turgid. They do not burst because of the rigid cell wall. If a plasmolysed plant cell is again placed in hypotonic solution, water will enter the cell. This process is called **deplasmolysis**. When plant cells are placed in **isotonic** solution, no change will occur because the amount of water entering and leaving the cell is at equilibrium.

Aim

To investigate how plasmolysis and deplasmolysis occur in onion and elodea cells when they are placed in solutions of different salt concentrations.

Equipment

- purple onions
- fresh elodea leaves
- distilled water
- 15% sodium chloride solution (NaCl)
- a pipette
- 2 microscope slides
- 2 cover slips
- light microscope
- microscope lamp
- forceps
- paper towel
- camel hair brush
- lens tissue
- scalpel

SAFETY RULES

- ⚠ Be careful when handling the microscope, slides and cover slips.
- ⚠ When viewing through high-power objective, take care not to break the cover slip by lowering the tube too fast. Use the fine focus adjustment to focus the object.
- ⚠ After finishing the experiment, replace the microscope in its box.
- ⚠ Clean the slides and cover slips with soapy water, dry them with paper towel and put them away in their boxes.
- ⚠ Do not discard any plant material into the sink.

Method

Part A – Observing onion cells

- 1 Cut an onion into small cubes. Remove a small section of tissue from the inner epidermis of the onion.
- 2 Put the onion epidermis on the microscope slide. Add one drop of distilled water and place a cover slip over the top. Push down on the cover slip with a piece of lens tissue to flatten the onion cells.
- 3 Observe the onion cells under the microscope at both low-power and high-power objectives. Under Results, draw a diagram of the onion cells and describe what you see.
- 4 Use the pipette to place several drops of the 15% NaCl solution under the cover slip. When adding the NaCl to the onion cell mount, place the top of the pipette next to the edge of the cover slip and apply a few drops. The NaCl solution will disperse under the cover slip to cover the onion cells.
- 5 Watch the interior of the onion cells for 2–3 minutes. Under Results, draw a diagram of plasmolysed onion cell and describe what was happening.
- 6 To remove the NaCl solution, place a dry paper towel along the edge of the cover slip. The capillary action of the water will draw the solution away from the onion cells.
- 7 To observe **deplasmolysis**, apply several drops of distilled water to the cover slip. Apply dry paper towel to the edge and draw up the solution. Reapply the distilled water.
- 8 Observe the interior of the onion cells for 2–3 minutes.
- 9 Under Results, draw a deplasmolysed onion cell and label the parts. Describe what was occurring.

Part B – Observing elodea cells

- 1 Obtain a leaf from the tip of an Elodea plant and repeat steps 2–7 in Part A. Observe the process of plasmolysis and deplasmolysis in elodea cells. Draw a plasmolysed and a deplasmolysed elodea cell and label the parts. Describe what was occurring in your Results.
- 2 Clean your equipment and carefully put the microscopes in their boxes.

Results

Draw diagrams in the table below and label the parts.

Onion cells before adding NaCl solution	Onion cells after adding 15% NaCl solution (hypertonic solution)	Onion cells after adding distilled water (hypotonic solution)
Elodea cell before adding NaCl solution	Elodea cells after adding 15% NaCl solution (hypertonic solution)	Elodea cells after adding distilled water (hypotonic solution)

Analysis of results

- 1 Describe the processes of plasmolysis and deplasmolysis.
- 2 Why does the cell shrink during plasmolysis?
- 3 How did the 15% salt solution affect the structure of the onion cell and the elodea cell? Explain why this occurred.
- 4 How did the distilled water affect the structure of the onion cell and elodea cell? Explain why this occurred.
- 5 Are plant cells normally hypertonic, hypotonic or isotonic to their environment? Why?
- 6 When do plant cells become turgid?
- 7 Can plant cells burst? Explain.
- 8 From your observations, is elodea a freshwater or saltwater plant? Explain.
- 9 In the winter, grass often dies near roads that have been covered in salt to remove the ice. What do you think is the reason for the grass dying?
- 10 Why do people who work in grocery stores spray fresh fruits and vegetables with water?
- 11 If a shipwrecked crew drink sea water they will probably die. Why?
- 12 If fresh strawberries are sprinkled with sugar, a few minutes later they will be covered with juice. Why?

Conclusion

- 1 In this experiment which solution is considered hypertonic? Which solution is considered hypotonic? Explain.

EFFECT OF SURFACE AREA TO VOLUME RATIO ON CELL SIZE



Background

Timing: 4 × 45 minute lessons

Most cells are microscopic in size because as a cell gets larger, it needs more food and produces more waste. Therefore, more materials must be able to move into and out of the cell through the **cell membrane**. To keep up with these demands, a growing cell needs a larger **surface area** through which to exchange materials. As the cell's **volume** increases, its outer surface grows too. However, the volume of a cell increases at a faster rate than the area of its outer surface. As the size of the cell increases the **surface area to volume ratio** decreases. Materials must be able to reach all parts of a cell quickly. When volume is too large relative to surface area, diffusion cannot occur at sufficiently high rates.

The shapes of cells can also affect the amount of surface area of a cell. Cells that are spherical in shape have a large surface area to volume ratio and are more efficient in obtaining nutrients, gases and getting rid of wastes. However, cells come in a great variety of shapes. Rod-shaped cells have more surface area compared to their volume than spherical cells.

In this experiment, the cell models are made of agar which has an acid-sensitive dye, **phenolphthalein**, incorporated into it. The dye changes from pink to clear in the presence of acid. The uptake of acid and hence colourless areas of the cell models represents the uptake of nutrients by the cell.

Aim

To use a model to investigate:

- The effect of size and shape on SA:V ratio.
- The effect of SA:V ratio on the rate of diffusion into a cell.

Hypothesis

Write a hypothesis to explain how surface area to volume ratio affects cell size, cell shape and a cell's ability to obtain nutrients.

Equipment (per group)

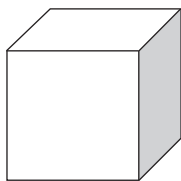
- | | | |
|--|---|-----------------|
| • 3 agar cubes measuring 1 cm ³ , 2 cm ³ and 3 cm ³ | • one cylinder with 1 cm diameter and 3 cm height | • plastic ruler |
| • one agar cube measuring 1 cm ³ | • two 250 mL beakers | • spatula |
| • one sphere with 1 cm diameter | • 0.1 M sulfuric acid (H ₂ SO ₄) | • glass rod |
| | | • knife |
| | | • paper towel |

SAFETY RULES

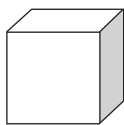
- ⚠ Handle the agar cubes carefully as they break easily. Use spatulas when taking them out of the acid.
- ⚠ If your skin comes in contact with the acid, wash your hands immediately with cold water.
- ⚠ Be careful when using knives.
- ⚠ Dispose of the agar cubes, paper towels and acid in specified containers.

Preparation of agar jelly

- 1 Add 9 g of agar to 400 mL of boiling water. Stir until the agar dissolves.
- 2 Dissolve 2 g of NaOH in 50 mL water. Add to the agar solution.
- 3 When the solution reaches 60°C, add phenolphthalein solution one drop at a time until the jelly turns pink.
- 4 Pour the solution into a rectangular cake pan about 3 cm deep and allow it to set.
- 5 After it is completely set, cut the following shapes from the jelly for each group. Cubes: 1 × 3 cm³; 1 × 2 cm³; 1 × 1 cm³. Shapes: 1 × cylinder with 1 cm diameter and 3 cm height; 1 × sphere with 1 cm diameter; 1 × cube 1 cm³.
- 6 To make 0.1 M sulfuric acid, add 10 mL concentrated acid to 1000 mL water. **Note:** Add the acid to the water, **not** the water to the acid.



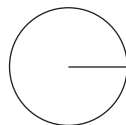
3 cm³ cube



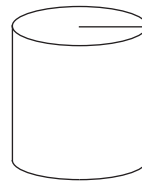
2 cm³ cube



1 cm³ cube



sphere
radius 0.5 cm



cylinder
radius 0.5 cm
height 3.0 cm



1 cm³ cube

Method

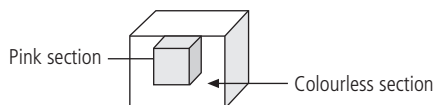
Part A – Effect of size

- 1 Before the laboratory exercise, use the following formulae to calculate the surface areas, volumes and surface area to volume ratios of the cubes and shapes. Enter the results in Table 1 and Table 3.

Formulae

Shape	Surface area	Volume
cube	$S = 6l^2$	$V = l^3$
sphere	$4\pi r^2$	$\pi r^2 h$
cylinder	$2\pi r(r + h)$	$\frac{4}{3}\pi r^3$

- 2 Divide the class into 6 groups of 3–4 students. Each group should take one cube of each size (3 cubes in total) measuring 3 cm³, 2 cm³ and 1 cm³ and place them on a paper towel.
- 3 Use a spatula to place three cubes into a 250 mL beaker. Cover the cubes with sulfuric acid (H₂SO₄), carefully turning them over with the spatula. Leave for 10 minutes.
- 4 After 10 minutes, remove the cubes from the acid, put them on a paper towel and blot them dry. Immediately cut each cube into two halves with a knife and measure the coloured part of the cubes.
Note: When the acid diffuses into the cubes, the pink colour of the cubes becomes colourless.



- 5 Record your results in Table 2 and calculate the percentage of uncoloured portion. This indicates the rate of diffusion.

Results

Table 1 Before diffusion

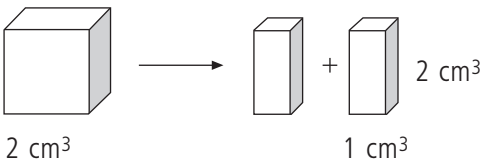
Cube	Surface area $S = 6l^2$	Volume $V = l^3$	Ratio SA:V	Reduced ratio
3 cm ³				
2 cm ³				
1 cm ³				

Table 2 After diffusion

Cube	Dimensions of coloured cube cm	Volume of coloured cube (A) cm ³	Volume of whole block (B) cm ³	Volume of uncoloured portion (B – A) cm ³	Percentage of uncoloured portion $(B - A) \times 100 \div B$
3 cm ³					
2 cm ³					
1 cm ³					

Analysis of results

- 1 Which cube has the greatest surface area? Volume? SA:V ratio?
- 2 What happens to the surface area as the cubes get larger? What happens to the volume as the cubes get larger?
- 3 What happens to the SA:V ratio as the cubes get larger? Proportionately, which grows faster – surface area or volume? Explain.
- 4 Explain what would happen to the volume, surface area and SA:V ratio of a cube measuring 2 cm³ if it is cut into 2 equal halves as shown in the following figure:



Part B – Effect of shape

Method

- 1 Calculate and record the surface areas, volumes and SA:V ratios of the sphere, cylinder and cube in Table 3.
- 2 Put the sphere, cube and cylinder in a beaker containing H₂SO₄ for 10 minutes.
- 3 Use a spatula to remove the shapes from the acid. Quickly dry them with paper towel.
- 4 Cut each shape in two halves with a knife. Measure the coloured parts of each shape.
- 5 Record your results in Table 4 and calculate the percentage of uncoloured portion. This indicates the rate of diffusion.

Results

Table 3 Before diffusion

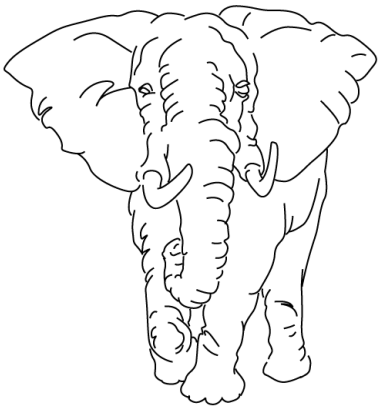
Cell shape	Surface area cm ²	Volume cm ³	Ratio SA:V	Reduced ratio
Cube $l = 1$ cm				
Sphere $r = 0.5$ cm				
Cylinder $r = 0.5$ cm $h = 3$ cm				

Table 4 After diffusion

Cell shape	Dimension of coloured portion (cm)	Volume of coloured portion (A) cm ³	Volume of whole block (B) cm ³	Volume of uncoloured portion (B – A) cm ³	Percentage of uncoloured portion $\frac{B - A}{B} \times 100$
Cube $l = 1 \text{ cm}$					
Sphere $r = 0.5$					
Cylinder $r = 0.5$ $h = 3$ $\pi = 3.14$					

Analysis of results

- 1 Which shape has the greatest surface area? Volume? SA:V ratio?
- 2 Which of the shapes you tested were the most efficient?
- 3 Which cell would be in greatest danger of starving? Explain.
- 4 Small animals like mice have a lot of skin on the outside compared to a small volume of bones, blood, and internal organs on the inside. Explain why they are voracious feeders and have a higher metabolic rate than large animals.
- 5 When an animal like an elephant has a large interior volume, the skin on the outside (the surface area) gets progressively smaller when it is compared to the volume inside. Explain why they have a lower metabolic rate and eat less than mice.



Conclusion

- 1 What types of shapes would you expect unicellular organisms to have in order to obtain nutrients and gases efficiently?
- 2 Why does a growing cell eventually have to split?
- 3 What happens to the time taken for a substance to diffuse to the centre of the cube as the size of that cube increases?
- 4 As a cell increases in size, what are the implications for it getting what it needs and getting rid of what it doesn't need?

FACTORS AFFECTING THE CELL MEMBRANE PERMEABILITY



Background

Timing: 2 × 45 minute lessons

A **membrane** is the essential protective covering of a cell. Membranes separate one cell from another. All living organisms are made up of cells; so membranes are essential to all life. The cell membrane is made up of a **phospholipid bilayer** in which **protein molecules** float. The membrane also contains **cholesterol** molecules, which give the membranes some rigidity and water resistance. Any factors that distort or alter proteins and lipid molecules affect the permeability of the membrane.

Beetroot cells contain a pigment called **betacyanin** in their vacuoles. A special membrane called the **tonoplast** surrounds the vacuoles. When beetroot cells are subjected to variation in pH, detergent, alcohol and high temperatures, the pigment will leak out because these factors alter the tonoplast and cell membrane structures. Variations in pH and high temperatures affect the protein molecules in the cell membrane. Detergent and alcohol solutions disrupt the phospholipid bilayer.

SAFETY RULES

- ⚠ Take care when using the cork borer and knife to cut the beetroot.
- ⚠ Be careful when handling hot liquids.
- ⚠ If a corrosive liquid splashes on your skin, immediately wash your hands with cold water.

Part A – Effect of pH, detergent and alcohol on the cell membrane

Aim To investigate how changes in pH, concentration of detergent solutions and alcohol affect the structure of the cell membrane of beetroot cells.

Hypothesis

Write a hypothesis to predict how different solutions and pH affect the structure of the cell membrane of beetroot cells.

- Equipment**
- 11 petri dishes
 - waterproof pen
 - paper towel
 - forceps
 - knife
 - beetroot with skin removed
 - cork borer ($d = 1\text{ cm}$)
 - paper towel
 - marking pen
 - solutions of pH 2, 4, 6, 8 and 10
 - solutions of 1%, 25% and 50% alcohol
 - solutions of 1% and 5% detergent
 - distilled water
 - colorimeter (if available)
 - electronic balance
 - ruler

Method

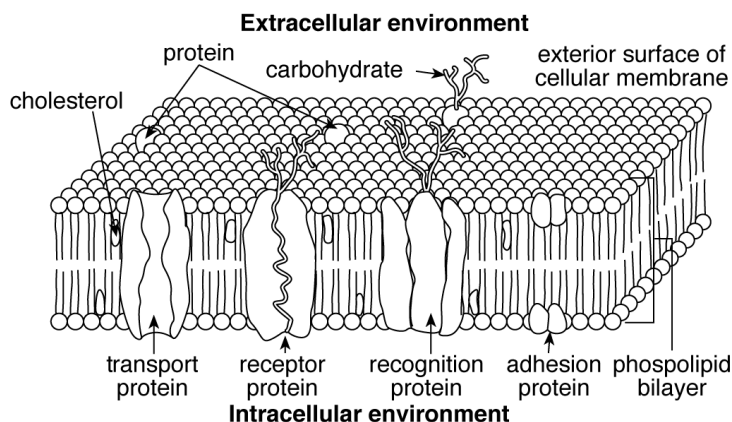
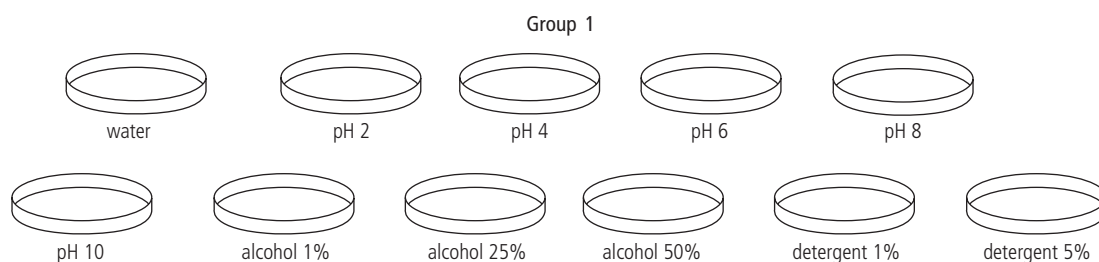


Figure 1: Cell membrane

Factors affecting the cell membrane permeability 6.2

- 1 Use a knife to cut off the top and bottom of a beetroot.
- 2 Push the cork borer into the beetroot and then withdraw it. The cylinder will remain inside the corer, so push it out with a borer of smaller diameter.
- 3 When a few uniform cylinders have been collected, cut them into pieces measuring 2 mm in length. The thinner cylinders will have a larger surface area to volume ratio and so the solutions will diffuse faster. Cut 55 cylinders.
- 4 Use an electronic balance to measure the mass of each cylinder. Make sure that all the cylinders are identical in length and mass.
- 5 Rinse the cylinders in cool running water for about 5 minutes or until the water runs clear.
- 6 Lay a paper towel on the bench and place 11 clean petri dishes on the paper.
- 7 Label the lids of the dishes 1 to 11. Label the paper towels with the name of the solutions and your group number as shown below.



Petri dishes										
1	2	3	4	5	6	7	8	9	10	11
Distilled water	pH 2	pH 4	pH 6	pH 8	pH 10	1%	25%	50%	1%	5%
Water	pH solutions					Alcohol			Detergent	

- 8 Use forceps to place 5 beetroot cylinders in each petri dish.
- 9 Pour the solutions into the appropriate petri dish so that the solution covers the beetroot cylinders. Place the lids on the dishes.
- 10 Observe the colour of the solution immediately after the set up, 1 hour after and 24 hours after. Record your results in Table 1.
- 11 If possible, use a colorimeter to measure the intensity of the colour of the solutions. If you do not have the colorimeter, record your results in Table 1 using a scale of 1–5:

1: very pale pink 2: pale pink 3: pink 4: red 5: dark red
- 12 Collate the class results and compare your results with the results of other groups.

Results

Table 1 Colour intensity of solutions over time

Time observed	Colour intensity of the solutions (arbitrary units)										
	Distilled water	pH					Alcohol			Detergent	
		pH 2	pH 4	pH 6	pH 8	pH 10	1%	25%	50%	1%	5%
At the start											
After 1 hr											
After 24 hrs											

Analysis of results

- 1 Which dish was used as the control? Did any changes occur in this dish?
- 2 What are the independent variables in this investigation?
- 3 What is the dependent variable? How would you measure it? What assumption do you make when choosing this method of measurement?
- 4 What other variables need to be controlled?
- 5 What is the ideal pH range for the beetroot cell membrane? Why do you think that the membrane leaked at pHs outside this range? (That is, which membrane component was affected by exposure to these pH solutions? Why?)
- 6 In which pH solution did the pigment leak out the most?
- 7 Which percentage of alcohol affected the membrane the most?
- 8 Which percentage of detergent affected the membrane the most?
- 9 Which solutions affected the membrane immediately?
- 10 From which part of the beetroot cell is the colour leaking out?
- 11 How did the beetroot cylinders appear after 24 hours in distilled water? Explain.

Part B - Effect of temperature on the cell membrane

Aim To determine the effect of temperature on the plasma membrane of a beetroot cell.

Hypothesis

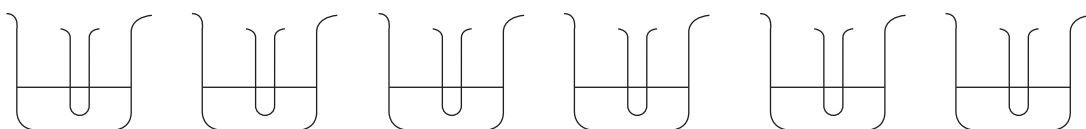
Write a hypothesis to predict how different temperatures affect the structure of the cell membrane.

Equipment

- cork borer ($d = 1 \text{ cm}$)
- beetroot (skin removed)
- knife
- white tile
- heat-proof mat
- Bunsen burner
- tripod
- electronic balance
- gauze mat
- 6 × 250 mL beakers
- 6 test tubes
- test tube holder
- Celsius thermometer
- distilled water
- tongs
- 10 mL measuring cylinder
- 2 test tube racks
- stopwatch
- colorimeter (if available)

Method

- 1 Using the cork borer, cut 66 beetroot cylinders measuring 2 mm in length. Use an electronic balance to weigh each cylinder to check that they have the same mass.
- 2 Put 10 ml of distilled water into each of 6 identical test tubes. Label the test tubes with your group name and as shown in the diagram below.



Room temp.	0°C	20°C	40°C	60°C	80°C
Control	1	2	3	4	5

- 3 Pour 200 mL water in each of 6 beakers and maintain them at room temperature, 0°C, 20°C, 40°C, 60°C and 80°C. Label the beakers control (room temperature) then 1 (0°C) to 5 (80°C).
- 4 Place the appropriately labelled test tube in its water bath. Allow 5 minutes for the water in the test tubes to reach the same temperature as the water baths.
- 5 When the water in the test tubes has reached the appropriate temperature, use forceps to place 5 beetroot cylinders into each test tube. Leave the test tubes in the water baths and maintain the temperatures for 30 minutes.

- 6 Remove the test tubes from the water baths. Shake each test tube to distribute the pigment evenly.
- 7 Remove the beetroot cylinders and place them in an empty beaker. Hold the test tubes to the light, and compare the colour of the liquid in each. Compare against the control (the test tube with distilled water at room temperature) to check for any variations in the colour of the water.
- 8 If possible, use a colorimeter to analyse the colour of the fluid in each test tube. If you do not have the colorimeter, record your results in Table 2 using a scale of 1–5.
 1: very pale pink 2: pale pink 3: pink 4: red 5: dark red
- 9 After the experiment, discard the beetroot slices in the bin provided, clean the equipment and pack up.

Results

Table 2

Temperature (°C)	Colour intensity of solutions (arbitrary units)
Control (room temp.)	0
0	1
20	2
40	3
60	4
80	5

Analysis of results

- 1 Use your results from Table 2 to draw a line graph.
- 2 Which test tube was used as your control?
- 3 During this experiment, how can you tell the degree to which the membrane has been damaged?
- 4 For which temperature does the membrane appear to suffer no damage?
- 5 At which temperature did the colour leak out the most?
- 6 Based on your knowledge of the two major structural components of the membrane, which component do you think would be most affected by temperature change? Explain what you think happened to this component to result in the membrane becoming leaky.
- 7 Is this reaction reversible? Explain.
- 8 Name some errors that could have occurred during this experiment. How could you improve the procedure to avoid these errors?

Conclusion

- 1 Describe what happened to the lipid layer and the protein molecules in the cell membranes when beetroot was placed in
 - a distilled water
 - b different pH solutions
 - c different strengths of alcohol
 - d different strengths of detergent.
- 2 Explain how different temperatures affected the structure of the cell membranes.
- 3 Did your experimental results support your hypothesis? Explain.

FACTORS AFFECTING CATALASE ACTIVITY



Background

Timing: 2 × 45 minute lessons

Enzymes are **globular proteins**. They act as **catalysts** that speed up chemical reactions without being destroyed or altered during the process. Enzymes are extremely efficient and may be reused. Enzymes are specific and react with a particular **substrate**. A substrate is a molecule with which the enzyme reacts and produces products. The **active site** is the only part of the enzyme that reacts with the substrate. However, its unique protein structure under certain circumstances can easily be **denatured**.

Some of the factors that affect enzyme reactions are enzyme concentration, pH, temperature, substrate and product concentration, activators and inhibitors. Enzymes react best at optimum temperature and optimum pH. If the environment of the enzyme is too acidic or too basic, the enzyme may irreversibly denature until it no longer has the necessary shape for proper functioning. The enzyme tested in this experiment is **catalase**, an intracellular enzyme. It is produced by cells to prevent the accumulation of toxic hydrogen peroxide. Hydrogen peroxide is decomposed by catalase to water and oxygen. This reaction occurs spontaneously without catalase, but the enzyme speeds the reaction considerably.

SAFETY RULES

- ⚠ Be careful when handling corrosive liquids. H_2O_2 is a strong oxidiser and will burn skin. Wash your hands immediately after any exposure.
- ⚠ Wear safety goggles, gloves and a lab coat to protect you from corrosive chemicals.
- ⚠ Dispose of the liver in a specified container. Do not wash the liquids down the drain.
- ⚠ Be careful when handling hot liquids and scalpels.
- ⚠ Use soap and warm water to clean the equipment and your hands.

Part A – Effect of catalase on hydrogen peroxide

Aim To investigate the effect of catalase from different animal and plant cells on hydrogen peroxide.

Hypothesis

Write a hypothesis to explain how catalase reacts with hydrogen peroxide.

- Equipment**
- fresh calf liver
 - chicken meat
 - apple
 - potato
 - scalpel
 - 7 mini test tubes
 - test tube rack
 - forceps
 - Bunsen burner
 - matches
 - tongs
 - heat-proof mat
 - wooden splints
 - tripod
 - wire gauze
 - mortar and pestle
 - 250 mL beaker
 - spatula
 - 100 mL measuring cylinder
 - 6% hydrogen peroxide (H_2O_2)
 - electronic balance

Method

- 1 Take 7 mini test tubes and label them A to G. Pour hydrogen peroxide solution to a depth of 1 cm into each test tube.
- 2 Add the following substances to the seven test tubes:
 - test tube A: sand only
 - test tube B: whole liver
 - test tube C: liver and sand ground together
 - test tube D: boiled liver
 - test tube E: potato
 - test tube F: chicken
 - test tube G: apple

- 3 Note the rate of reaction by looking at the amount of bubbles produced. Use the following scale to assess the rate of reaction:

0	no reaction
1	very slow reaction
2	slow reaction
3	medium reaction
4	fast reaction
5	very fast reaction

- 4 Use a glowing splint to test the gas given off. (If it is oxygen the splint will continue to glow.) Record your results in Table 1.

Results

Table 1 Catalase reactions with hydrogen peroxide

Test tube	Materials used	Rate of reaction
A (control)	sand + H ₂ O ₂	
B	whole liver + H ₂ O ₂	
C	ground liver + H ₂ O ₂	
D	boiled liver + H ₂ O ₂	
E	potato + H ₂ O ₂	
F	chicken + H ₂ O ₂	
G	apple + H ₂ O ₂	

Analysis of results

- 1 What was the purpose of using only sand in test tube A?
- 2 How did the activity of ground liver compare with that observed for the whole liver? Explain.
- 3 What effect does boiling the liver have on catalase activity?
- 4 What happened to the hydrogen peroxide when it reacted with catalase enzyme?
- 5 Did all the animal cells and plant cells tested contain catalase? What evidence did you find that they did contain catalase?

Part B - Factors affecting catalase activity

Aim To design an experiment to investigate one of the factors affecting the rate of catalase reaction.

Hypothesis

Write a relevant hypothesis related to the factor you are investigating.

Equipment

Devise an experiment and choose the apparatus needed for your experiment.

Method

- 1 Divide into 4 groups. Each group should devise an experiment to investigate the effect of one of the four factors listed on the rate of catalase reaction.

- temperature
- pH
- substrate (H_2O_2) concentration
- enzyme (catalase) concentration

Note: Use the tables given below as a guide to planning experiments. Alternatively, devise your own. Write a report using the headings: Aim, Hypothesis, Equipment, Method and Results. Answer the questions given below for Analysing the results and the Conclusion.

Hints: Use whole liver measuring 1 g for the first 3 experiments. Use an electronic balance to measure the liver pieces accurately. Use the same scale as for Part A, shown below.

0: No reaction; 1: Very slow reaction; 2: Slow reaction; 3: Medium reaction; 4: Fast reaction; 5: Very fast reaction.

Results

Effect of temperature on enzyme activity

Hints: Use the same amount of liver and H_2O_2 but vary the temperature by using water baths ranging from 0°C to 100°C . For temperatures higher than 40°C , cool the liver before adding H_2O_2 because heat can decompose H_2O_2 .

Table 1

Temperature	Enzyme activity
0°C	
20°C	
40°C	
60°C	
80°C	
100°C	

Effect of substrate concentration on enzyme activity

Hints: Vary the amount of substrate but maintain the temperature, the pH and the quantity of liver as constants.

Table 3

Amount of substrate H_2O_2 (mL)	Enzyme activity
5	
10	
15	
20	
25	
30	
35	

Effect of pH on enzyme activity

Hints: Use the same amount of pH solutions and same amount of H_2O_2 and keep the temperature constant. Only vary the pH of the solutions.

Table 2

pH	Enzyme activity
1	
3	
5	
7	
9	
11	
13	

Effect of enzyme concentration on enzyme activity

Hints: Vary the amount of liver but maintain as constants the temperature, pH and H_2O_2 .

Table 4

Amount of liver catalase (g)	Enzyme activity
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	

Graphing

Using the class results, draw four line graphs to show the effect of temperature, pH, substrate concentration and enzyme concentration on catalase activity.

Analysis of results**Effect of temperature**

- 1 At which temperature did the enzyme react most vigorously? Least vigorously?
- 2 What was the optimum temperature for catalase?
- 3 What happened to the rate of reaction
 - at lower temperatures (0°C to 20°C)?
 - at higher temperatures (60°C to 100°C)?
- 4 Why did the reaction stop at higher temperatures? What happened to the enzyme?

Effect of pH

- 5 At which pH the enzyme reacted most vigorously? Least vigorously?
- 6 What was the optimum pH for Catalase enzyme?
- 7 What happened to the rate of reaction
 - below pH 7?
 - above pH 7?
- 8 Explain what happened to the enzyme at pH levels above and below 7.

Effect of substrate concentration:

- 9 What happened to the rate of reaction as the amount of H_2O_2 increased?
- 10 Explain why the rate of reaction levelled off.

Effect of enzyme concentration

- 11 How did the enzyme concentration affect the rate of reaction?
- 12 Is it possible to reuse the enzyme after a reaction? Explain.

Conclusion

- 1 What happened when hydrogen peroxide was added to whole liver, ground liver and boiled liver?
- 2 What is the main function of catalase enzyme in living cells? Write an equation.
- 3 What are the optimum temperature and pH for catalase activity?
- 4 How does the increase in the amount of substrate (H_2O_2) affect the rate of reaction?
- 5 How does the increase in the amount of enzyme (liver) affect the rate of reaction?
- 6 How could you improve your method to minimise or exclude experimental errors?

FACTORS AFFECTING THE RATE OF PHOTOSYNTHESIS



Background

Timing: 2 × 45 minute lessons

Photosynthesis is the process by which green plants use sunlight to produce glucose and oxygen from simple compounds such as water and carbon dioxide. Photosynthesis occurs in the **mesophyll** layer of leaves, which contains tiny bodies called **chloroplasts**. Chloroplasts contain a green pigment called **chlorophyll**, which catches light energy and converts it into chemical energy. Light intensity, carbon dioxide concentration, water and temperature affect the rate of photosynthesis. In this experiment, the effect of light intensity and carbon dioxide on the rate of photosynthesis and starch production in the elodea plant and the geranium plant will be investigated.

SAFETY RULES

- ⚠ Be careful when boiling leaves in alcohol because the fumes can catch fire. If the fumes do catch fire, turn off the Bunsen burner and smother the flame with sand from the fire bucket. Do not blow the flame out. Use a fire blanket if clothing catches fire.
- ⚠ Make sure you turn off the Bunsen burner after the water has boiled.
- ⚠ Dispose of hot alcohol in a specified container.
- ⚠ Use gloves and goggles when handling sodium hydroxide because it is caustic. Wash your hands immediately with cold water if it spills on your hands.
- ⚠ Be careful when cutting the elodea plant. Cut it under water.

Part A - The effect of light intensity on photosynthesis

Aim To investigate the effect of light intensity on the rate of photosynthesis in the elodea plant.

Hypothesis

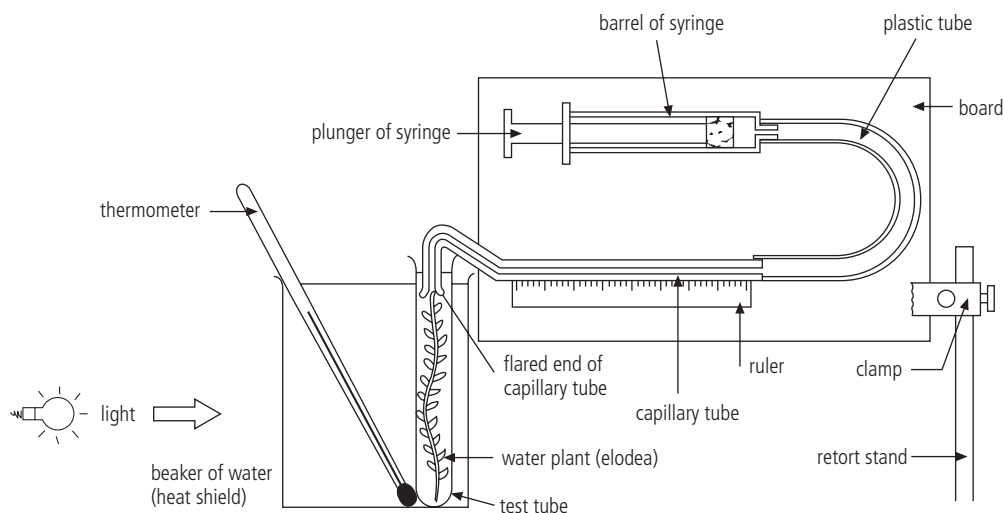
Write a hypothesis to predict how light intensity affects the rate of photosynthesis.

Equipment

- | | | |
|-------------------------------------|---------------------------|------------------|
| • elodea plant (<i>anacharis</i>) | • stop watch | • plastic tube |
| • 40-watt lamp | • large test tube | • thermometer |
| • test tube | • 500 mL beaker | • a wooden board |
| • scalpel | • syringe | • forceps |
| • distilled water | • retort stand with clamp | • light meter |
| • sodium bicarbonate of soda | • capillary tube | • metric ruler |

Method

Audus apparatus



- 1 Take an elodea plant measuring 3 cm in length. Remove a few leaves from the bottom of the stem. With the plant under water, cut the stem at an angle and use forceps to lightly crush the cut end.
- 2 Put 0.1 g bicarbonate of soda in a test tube filled with distilled water. Insert the elodea plant into the test tube, with stem end up.
- 3 Place the test tube in a beaker containing distilled water. This will act as heat shield.
- 4 Set up the apparatus as shown in the diagram above.
- 5 Place a 40-watt lamp 5 cm from the beaker. Close the blinds so that the room is dark. Use a light meter in lux to measure the light intensity. **Note:** Make sure the distance is measured from the middle of the plant to the filament of the globe.
- 6 Place a thermometer in the beaker. Measure the temperature of the water in the beaker at regular intervals. Maintain a constant temperature of the water throughout the experiment.
- 7 Switch on the lamp and wait for five minutes. Look for bubbles emerging from the cut end of the stem. If bubbles fail to appear, cut off more of the stem and crush the cut end with forceps.
- 8 Start a stop watch. After 5 minutes, carefully withdraw the bubble into the capillary tube and measure and record its length. The production of oxygen can be used as a measure of the rate of photosynthesis.
- 9 Repeat steps 5 to 8 with the lamp at distances of 10 cm, 15 cm, 20 cm, 25 cm, 30 cm, 35 cm, 40 cm and 45 cm.
- 10 Complete a second trial for each distance and find the average length. Calculate the volume of oxygen released in Table 1 by using the formula given.
- 11 Collect and average the class results. Draw a line graph plotting the average volume of oxygen on the vertical axis and the light intensity on the horizontal axis.

Results

Table 1 Effect of light intensity

Distance of light source (cms)	Light intensity (lux)	Oxygen released			
		1 st trial length (cms)	2 nd trial length (cms)	Average length (cms)	Volume ($3.14 \times \text{length} \times 0.1^2$)
5					
10					
15					
20					
25					
30					
35					
40					
45					

Analysis of results

- 1 How did this investigation demonstrate that plants give off oxygen during photosynthesis? Explain your answer based on your observations.
- 2 How did light intensity affect the rate of photosynthesis?
- 3 How was CO₂ supplied to the plant?
- 4 What were the sources of errors? How could you improve your method?



Part B – The effect of carbon dioxide concentration on starch production

Aim To investigate the effect of carbon dioxide concentration on starch production in geranium leaves.

Hypothesis

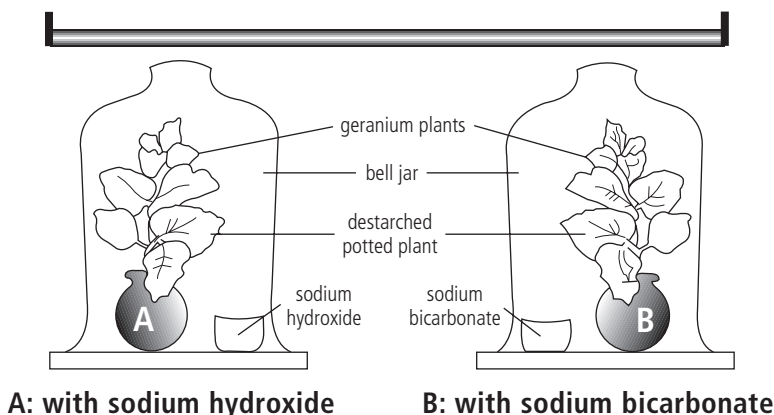
Write a hypothesis to predict how carbon dioxide concentration affects the rate of starch production.

Equipment

- 2 geranium plants the same size
- 2 bell jars
- 2 × 250 mL beakers
- 500 mL beaker
- 2 test tubes
- 2 petri dishes
- forceps
- alcohol
- Bunsen burner or hot plate
- tripod
- gauze mat
- heat mat
- tongs
- sodium bicarbonate
- sodium hydroxide
- distilled water
- paper towel
- petroleum jelly
- iodine
- fluorescent lamp
- pipette

Method

This part of the experiment can be done as a demonstration. Students may carry out the starch test.



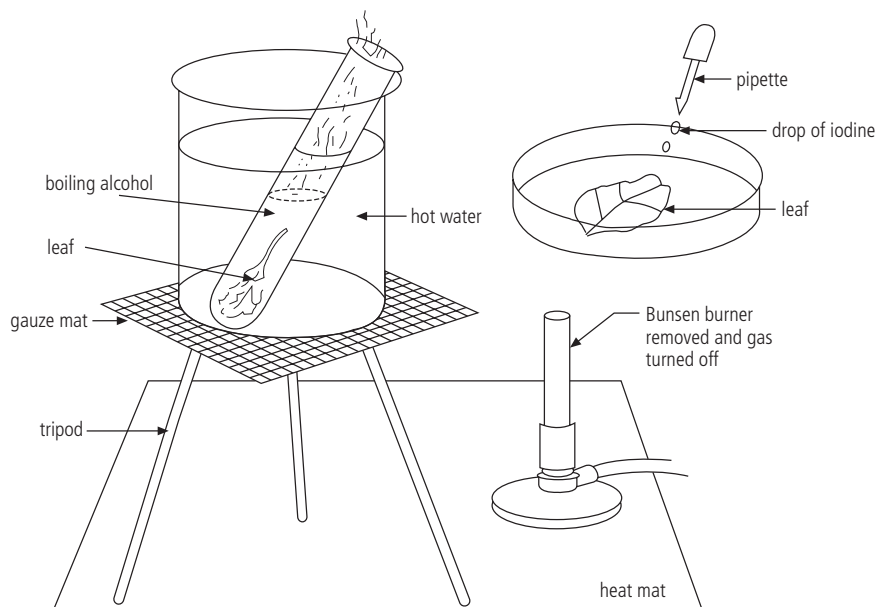
- 1 Mark two geranium plants as A and B.
- 2 Place them in a dark cupboard for 24 hours to destarch them. (Plants cannot make starch in the absence of light).
- 3 The next day, water both plants with the same amount of water.
- 4 Place a beaker half-filled with sodium hydroxide next to plant A and cover it with a bell jar. Use this set up as the control because sodium hydroxide absorbs CO_2 and so no CO_2 is available to plant A.
- 5 Place a beaker half filled with sodium bicarbonate next to plant B and cover it with a bell jar. Use this set up as the experimental variable because sodium bicarbonate releases CO_2 for the plant.
- 6 Place a fluorescent light above the bell jars so that both plants receive the same light intensity and heat. Make sure the room is darkened.
- 7 Leave the set up for 24 hours. Then pick a leaf from each plant and test for the presence of starch.

Starch test

Students may do this part of the experiment.

- 8 Half fill a beaker with hot water and use a Bunsen burner to bring it to the boil.
- 9 Drop a leaf from plant A into the boiling water for 5 seconds. This will kill the cells and break down the cell walls and membranes so that the iodine solution can penetrate.
- 10 Use forceps to remove the leaf from the boiling water. Place it in a test tube half filled with alcohol. Stand the test tube in the beaker containing boiling water. The alcohol will start to boil and turn green in colour because the chlorophyll in the leaf is soluble in alcohol.

- 11 After 10 minutes the leaf should look pale in colour. If it did not turn pale, place the test tube containing the alcohol and the leaf in a water bath on an electric hot plate for a few minutes.
- 12 Remove the leaf with forceps and dip it in hot water to soften it. Place the leaf on paper towel to dry, and then place it in a petri dish. **Note:** Be careful while heating alcohol because the fumes of alcohol can catch fire (see Safety Rules).



- 13 Put several drops of **potassium iodide** onto the leaf and let it stand for a few minutes. Then drain the excess into a beaker.
- 14 Observe the colour change in the leaf and record your results in Table 2. (The darker the purple colour, the more starch is present. The lighter the colour, the less starch is present.)
- 15 Repeat steps 8 to 14 with the leaf from plant B and record your results.
- 16 Dispose of the iodine and alcohol as specified by your teacher, and clean up and put away all the materials. Discard the leaves in a rubbish bin.

Results

Table 2 Effect of carbon dioxide on starch production

Plants	Colour of leaf after adding potassium iodide	Amount of starch present
Control plant A: no CO ₂		
Experimental plant B: with CO ₂		

Analysis of results

- 1 How did the presence of sodium hydroxide affect the rate of photosynthesis and starch production in plant A?
- 2 How did the presence of sodium bicarbonate affect the rate of photosynthesis and starch production in plant B?
- 3 What were the sources of error in this experiment? How could they be rectified?

Conclusion

- 1 Define photosynthesis. Write a balanced equation for photosynthesis.
- 2 Explain how light intensity and carbon dioxide affected the rate of photosynthesis and starch production in this experiment.
- 3 Did your results support your hypothesis?

AEROBIC AND ANAEROBIC RESPIRATION



Background

Timing: 2 × 45 minute lessons

Cellular respiration is the **oxidation** of glucose to release energy. There are two types of cellular respiration. **Aerobic** respiration occurs in the presence of oxygen. The glucose is completely broken down to carbon dioxide and water. **Glycolysis** is the initial stage when the glucose molecule is split into two **pyruvate** molecules and two **ATP** molecules are released. **Anaerobic** respiration takes place in the absence of oxygen. **Fermentation** is the process when pyruvate is converted to lactic acid in animals. When fungi and yeast respire anaerobically they produce ethanol (alcohol) and carbon dioxide. In this experiment, aerobic respiration of germinating pea seeds and anaerobic respiration of yeast cells will be investigated.

SAFETY RULES

- ⚠ Wear goggles and a lab coat to protect from accidental splashes.
- ⚠ Do not taste any of the experimental solutions.
- ⚠ Do not pour the solutions down the drain after the experiment.
- ⚠ Dispose of the solutions and seeds in specified containers.
- ⚠ Before germinating the pea seeds, soak them in formalin to remove bacteria and fungus.

Part A – Aerobic respiration by germinating peas

This part of the experiment can be set up as a demonstration for one week.

Aim To observe the release of heat energy and the formation of carbon dioxide by germinating pea seeds in the presence of oxygen.

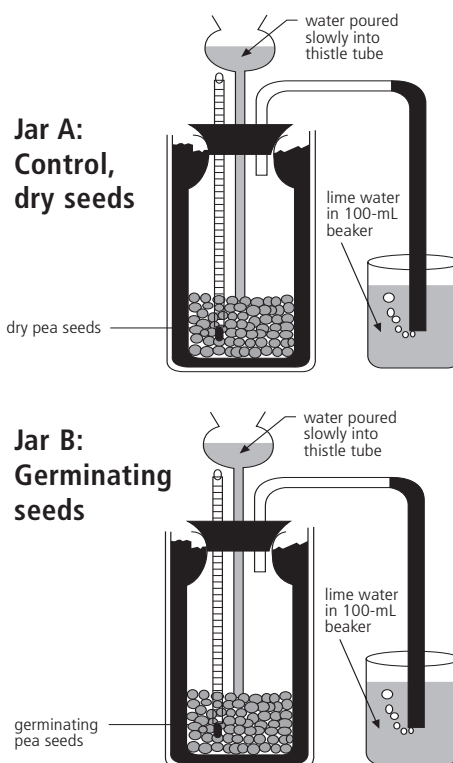
Hypothesis Write a hypothesis to predict what will happen during the aerobic respiration of pea seeds.

- Equipment**
- 2 gas jars
 - 2 × three-holed rubber stoppers
 - 2 thermometers
 - 2 thistle funnels
 - 2 double-bent glass tubes
 - vaseline
 - 2 rubber tubings
 - 2 pinch clamps
 - 2 metal cans to hold jars
 - sawdust
 - 2 × 250 mL beakers
 - tap water (room temperature)
 - 100 pea seeds
 - lime water

Method

Note: Pea seeds can be germinated in advance to reduce the time of observation.

- 1 Set up the apparatus as shown in the diagram above. Label the jars A and B.
- 2 Put some sawdust in the bottom of two cans. Place one gas jar in the centre of each can, on the sawdust. Add sawdust around the jars. The sawdust insulates the apparatus like a calorimeter.
- 3 Put 50 dry pea seeds in each jar. Add tap water to jar B until the seeds are just covered. Do not add water to jar A as it will be used as the control.
- 4 Place thermometers in jar A and jar B so the tips of the thermometers are in the centre of the seeds.
- 5 Secure the pinch clamp at the end of the rubber tubing for jar A and jar B to seal the air.



- 6 Check the temperature of both jars daily for 5 days. Record your results in Table 1. If you completely remove the thermometer, always put your finger over the hole of the stopper to keep the air inside the container from mixing with the outside air.
- 7 After the seeds in jar B have germinated, place a beaker filled with lime water next to each jar. Remove the pinch clamps from the rubber tubing and place the tubing into the lime water.
- 8 Pour water into the thistle funnels into jars A and B until each jar is filled. This will force CO₂ out of the jars. Note the colour of the lime water and record your results in Table 1. (If the lime water turns milky, carbon dioxide is present.)

Results

Table 1 Testing for temperature rise and carbon dioxide

Time of measurement	Temperature reading (°C)		CO ₂ release Colour of lime water	
	Jar A Dry seeds	Jar B Germinating seeds	Jar A Dry seeds	Jar B Germinating seeds
Day 1				
Day 2				
Day 3				
Day 4				
Day 5				

Analysis of results

- 1 Why were the jars packed in sawdust?
- 2 Which jar was used as the control?
- 3 What were the independent and dependent variables?
- 4 From the data collected in your observations, what evidence was there that germinating peas are releasing heat energy?
- 5 What gas was produced during the aerobic respiration of germinating seeds?
- 6 What did you observe as the gas from the jar A and jar B bubbled through the lime water?

Part B – Anaerobic respiration by yeast cells

Aim To observe the release of heat energy and the formation of carbon dioxide and ethanol by yeast cells in the absence of oxygen.

Hypothesis Write a hypothesis to predict what will happen during the anaerobic respiration of yeast cells.

Equipment

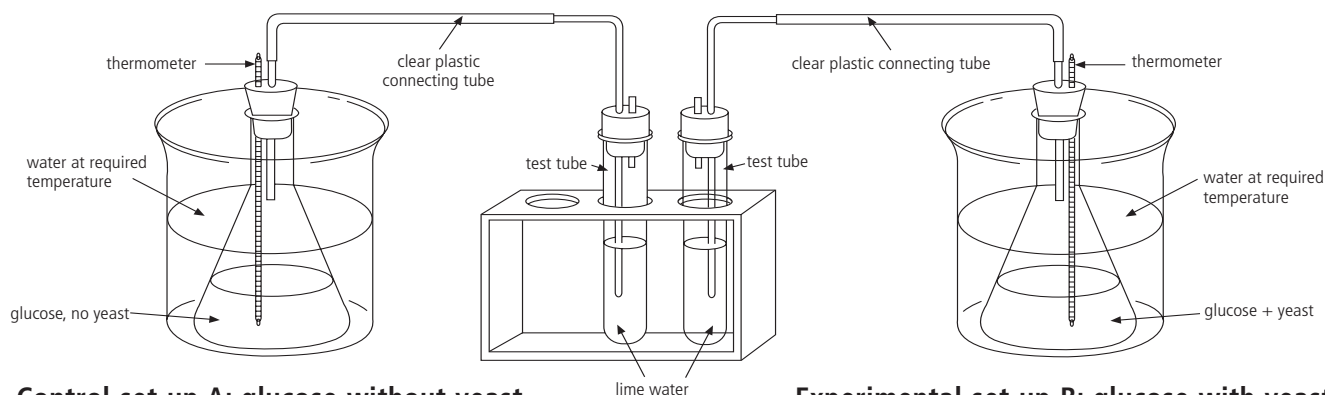
- 2 × 100 ml conical flasks
- 2 × two-holed rubber stoppers
- 2 straight glass tubings
- 2 double-bent glass tubings
- 2 plastic tubes
- 100 mL measuring cylinder
- glass stirring rod
- 2 g compressed dry yeast
- 2 thermometers
- lime water
- 2 test tube racks
- 2 test tubes with one-holed rubber stoppers
- 20% glucose solution
- hot plate
- 2 × 500 mL beakers
- vaseline

Advance preparation

- If the lime water is cloudy, filter it before use.
- Add 20 g of glucose to 80 mL of tap water to make 20% glucose solution. The day before the experiment, boil and cool the glucose solution to remove oxygen.
- Weigh out 2 g portions of 1 cm³ yeast pieces. Compressed yeast is better than dried yeast.

Method

- 1 Set up the apparatus as shown. Label one conical flask A and the other B. Pour 25 mL of glucose solution into each conical flask. Do not add yeast to the control flask A.

**Control set up A: glucose without yeast****Experimental set up B: glucose with yeast**

- 2 Add 2 g compressed yeast to the glucose solution in flask B. Stir with a glass rod until the yeast has dissolved. (Yeast needs 3–5 min to begin active respiration.)
 - 3 Add 200 mL of hot water to each of 2 beakers. Maintain the temperatures of the beakers at 40°C by adding hot water or cold water.
 - 4 Place a thermometer and a delivery tube through the stopper of each flask. Apply vaseline around the openings to prevent the entry of air.
 - 5 Place each test tube in a test tube rack. Pour 4 mL of lime water into each test tube. Make sure the stoppers fit tightly and apply vaseline.
 - 6 At 10-minute intervals, note the temperature and the colour of the lime water. Record your results in Table 2 for readings at 10, 20, 30, 40, 50 and 60 minutes.
 - 7 Collate the class results. Use the data to draw a line graph. Label the x-axis: Time in minutes. Label the y-axis: Temperature in degree Celsius.
 - 8 After 60 minutes, open the flasks and note the smell and colour changes in the glucose solution.
- Note:** All the fittings must be airtight to create the anaerobic condition.

Results

Table 2

Time (minutes)	Temperature (°C)		Release of CO ₂ Colour of lime water		Changes in glucose solution	
	Control Flask A	Experimental Flask B	Control Flask A	Experimental Flask B	Control Flask A	Experimental Flask B
0						
10						
20						
30						
40						
50						
60						

Analysis of results

- 1 What was the control? Why was it necessary to use a control?
- 2 What were the independent and dependent variables?
- 3 How do you know that fermentation is occurring?
- 4 What changes occurred in the temperature and the lime water after 60 minutes?
- 5 How did you infer that CO₂ has been released?
- 6 What changes have occurred to the glucose solution in the control flask and the experimental flask?
- 7 How could you improve the experimental set up in order to rectify experimental errors?

Conclusion

- 1 Summarise what occurred during the aerobic respiration of germinating pea seeds and the anaerobic respiration of yeast cells.

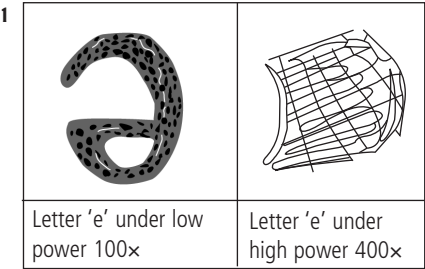
SUGGESTED ANSWERS

1 Introduction to light microscope

Calculation of total magnification

Total magnification: Low power 10 × 5 = 50×; Low power 10 × 10 = 100×; High power 40 × 10 = 400×

Results



2 Size of one paramecium = 1600 ÷ 8 = 200 microns

3

Eyeiece	Objective	Magnification (eyeiece × objective)	Field of view (in microns)	No. of paramecia	Size (in microns)
10×	10×	10 × 10 = 100×	1600	8	200
10×	40×	10 × 40 = 400×	400	2	200

Analysis of results

1

Part of microscope	Function	Part of microscope	Function
Eyeiece	A lens through which you see the object.	Mirror	Directs light upwards onto the slide.
Body tube	Supports the eyeiece.	Coarse-focus adjustment	Moves the body tube up and down, making large adjustments to the focus.
Revolving nosepiece	Holds the objectives.	Fine-focus adjustment	Makes small adjustments to the focus.
Low-power objective	A small lens with low magnifying power.	Arm	Attaches the eyeiece and body tube to the base.
High-power objective	A large lens with high magnifying power.	Stage clips	Metal clips that hold a slide securely onto the stage.
Stage	The platform on which a slide is placed.	Inclination joint	An adjustable joint that lets the arm tilt at various angles.
Diaphragm	Allows different amounts of light onto the stage.	Base	Supports the microscope.

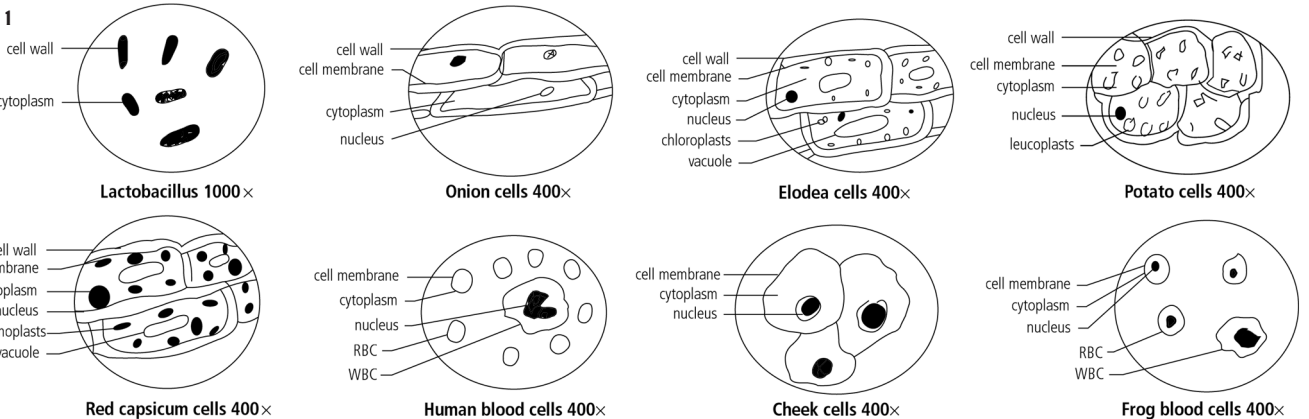
2 a To the right: The image moved to the left. b To the left: The image moved to the right. c Up: The image moved down.
d Down: The image moved up.

Conclusion

1 a Enlargement of image. b Amount of object seen under the microscope. c The ability to distinguish two objects that are close together as separate objects. 2 Upside down and back to front (laterally reversed). Under low power, the whole letter can be seen. Under high power, only a part of the letter can be seen but it was magnified 4 times. 3 a size of the organism = 2000 microns ÷ 4 = 500 microns
b The high power field of view = 2000 ÷ 4 = 500 microns c Cells seen under high power = 4 ÷ 4 = 1 cell

2 Observing cells

Results



2 Observing cells (cont'd)

Table 1 – Organelles seen in prokaryotic cells and eukaryotic cells

Cell organelles	Prokaryotic cells	Eukaryotic plant cells					Eukaryotic animal cells	
	Lacto-bacillus	Onion cells	Elodea cells	Potato cells	Red pepper cells	Cheek cells	Human RBC	Frog RBC
Cell wall	✓	✓	✓	✓	✓	✗	✗	✗
Cell membrane	✗	✓	✓	✓	✓	✓	✓	✓
Cytoplasm	✓	✓	✓	✓	✓	✓	✓	✓
Nucleus	✗	✓	✓	✓	✓	✓	✗	✓
Mitochondria	✗	✗	✗	✗	✗	✗	✗	✗
Chloroplasts	✗	✗	✓	✗	✗	✗	✗	✗
Leucoplasts	✗	✗	✗	✓	✗	✗	✗	✗
Chromoplasts	✗	✗	✗	✗	✓	✗	✗	✗
Vacuole	✗	✓	✓	✓	✓	✗	✗	✗

Table 2 – Differences between plant and animal cells

Cell organelle	Plant cell	Animal cell
Cell wall	✓	✗
Chloroplasts	✓	✗
Vacuole	✓	✗

Analysis of results

1 Lactobacillus is rod shaped. **2** Cell wall and cytoplasm **3** Prokaryotic cells have no membrane-bound cell organelles. They do not possess a membrane-bound nucleus. Eukaryotic cells have numerous membrane-bound cell organelles and possess a membrane-bound nucleus.

Onion cells **4** Rectangular in shape. **5** Stained with potassium iodide: cell wall, cell membrane, cytoplasm and nucleus. Stained with Janus Green B: Mitochondria. **Elodea cells** **6** Stained with potassium iodide: cell wall, cell membrane, cytoplasm, chloroplasts, vacuole and nucleus.

7 chloroplasts were seen around a large vacuole. When warm water was added, cytoplasmic streaming was seen. Nutrients and cell products are carried to different parts of the cell by this process. **8** Cytoplasmic streaming stopped because the iodine was toxic and killed the cells. **Potato cells** **9** Leucoplast which stores starch. **10** Leucoplasts turned purple when iodine was added. **Red capsicum cells** **11** Chromoplasts. **12** Chromoplasts stored coloured pigments which give colour to the cells. **Cheek epithelial cells** **13** Irregular in shape; they protect the underlying cells in the mouth. **14** Cell membrane, cytoplasm and nucleus **Human and frog red blood cells** **15** Human RBC have no nuclei and frog RBC have nuclei. **16** White blood cells have large nuclei and are irregular in shape. RBC are disc shaped and have no nuclei. **17** Plant cells have a cell wall and large vacuoles. Some plant cells have chloroplasts, some have leucoplasts and others have chromoplasts.

18	Stain added	Effect of the stain on cells
	Crystal violet added to lactobacillus	Only cell wall and cytoplasm were stained purple.
	Potassium iodide added to onion, elodea and potato cells	Cell wall, cell membrane and nucleus were stained brown, chloroplasts and leucoplasts were stained purple. Cytoplasmic streaming in elodea stopped.
	Janus green B stain added to onion cells	Mitochondria were stained blue.
	Methylene blue added to cheek and blood cells	Cell membrane, cytoplasm and nucleus were stained blue.

Conclusion

1 Cells have different structures and possess different cell organelles because different cells have different functions. Their shapes are adapted to carry out specific functions. **2** If the cell has a cell wall and a large vacuole and is not moving, it would be identified as a plant cell. If it is small, irregular in shape, has no cell wall and is moving, it would be identified as an animal cell. **3** Prokaryotic cells have a cell wall, cytoplasm, ribosomes and a circular chromosome which is not enclosed in a nuclear membrane. Eukaryotic cells have more cell organelles which are membrane bound. Nucleus contains linear chromosomes.



Diffusion and osmosis

Part A – Results

Table 1 – Starch test

Level of water results could vary

Experimental set up	Colour of water in the jar before and after diffusion	Colour of starch in the tubing before and after diffusion	Level of water in gas jar before and after diffusion	Level of water in the thistle funnel before and after diffusion	Interpretation
Jar A	Yellow to light yellow	White to blue-black	Slightly decreased	Slightly increased	Iodine diffused into the tubing and changed the colour of the starch

Table 2 – Glucose test

Level of water results could vary

Experimental set up	Colour of tes-tape in the gas jar before and after diffusion	Colour of tes-tape in the gas jar before and after diffusion	Level of water in gas jar before and after diffusion	Level of water in the thistle funnel before and after diffusion	Interpretation
Jar B	Yellow to green	Dark green to light green	Slightly increased	Slightly decreased	Glucose diffused out of the tubing into the water in gas jar and gave a positive test with tes-tape

Analysis of results

1 Starch solution colour changed to blue-black since iodine diffused into the cellophane tubing. 2 Iodine and water molecules could diffuse through the cellophane tubing because they are small, but starch molecules did not diffuse because they are larger in size. 3 Glucose molecules diffused through the dialysis tubing into the water in gas jar B. When tested with tes-tape, it turned green showing the presence of glucose in water. 4 In jar A, the starch level in the thistle funnel increased indicating that water from jar A has moved into the dialysis tubing. In the case of jar B, the glucose level in the thistle funnel decreased indicating that glucose and water moved out of the dialysis tubing into the gas jar B.

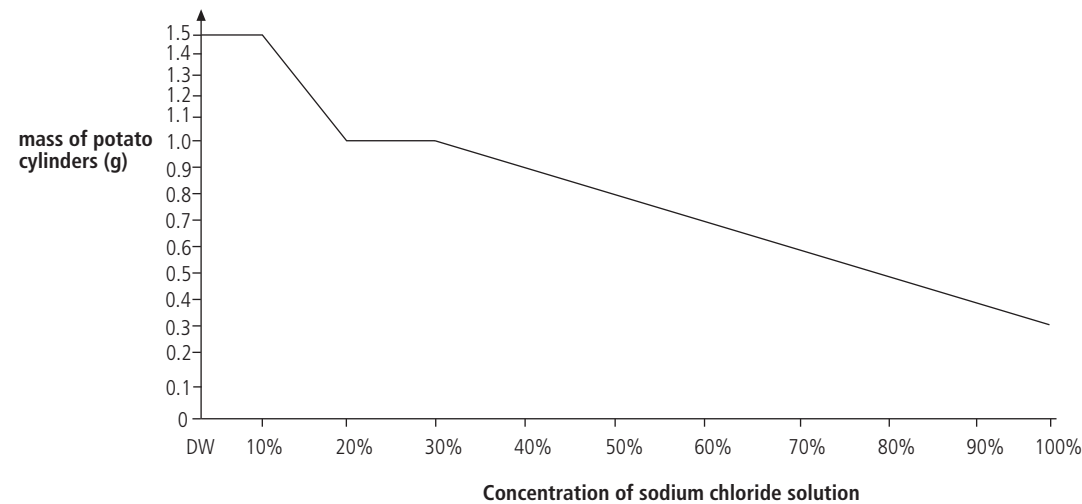
Part B

Results of table 3

Changes in length and mass of potato cylinders: Increased in distilled water and 10% salt solution, no change in 20% and 30% salt solutions, decreased in 40 %–100% salt solutions.

Graphing

Changes in mass of potato cylinders in different salt solutions



3 Diffusion and osmosis (cont'd)

Analysis of results

1 Test tube 1 with distilled water was used as the control. **2** The dependent variable was change in salt solution concentrations and the independent variable was the change in length and mass of the potato cylinders. **3** In more concentrated salt solutions, the water from potato cylinders moved out of the cell causing them to shrink and reduce in length and mass. **4** Medium strength salt solutions caused the smallest change because the solutions were isotonic and the amount of water moving in and out of the cells was equal. **5** No change would occur in the length and mass of the potato cylinders in 15% cell sap because the concentration of the salt inside and outside the cell is the same (isotonic). **6** Possible errors are: (a) The length and mass of the potato cylinders are not identical. (b) The salt solutions are not made up accurately (c) errors in measurement (d) the potato cylinders were not left in the salt solutions long enough.

Conclusion

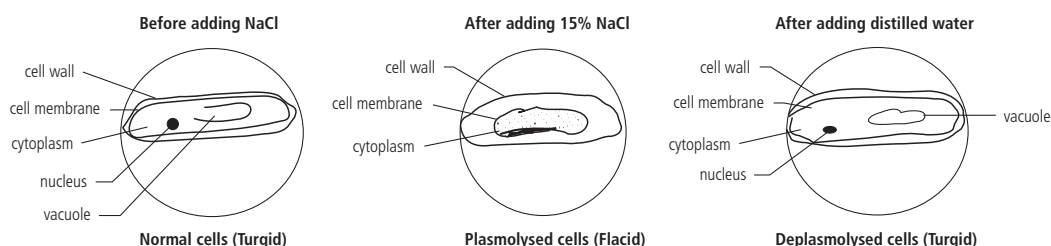
1 Diffusion is the movement of solids, gases and ions from a region of higher concentration to a region of lower concentration. Osmosis is the passage of water from a region of high water concentration to a region of low water concentration through a semipermeable membrane. **2** Iodine, water and glucose molecules diffused through the cellophane tubing. **3** In distilled water and low salt concentrations the length and mass of potato cylinders increased as water moved into the cells. In medium salt concentrations, no change in length and mass of the potato cylinders because the rate of water moving in was the same as the rate of water moving out of the cells. In high salt concentrations, length and mass of potato cylinders decreased because water moved out of the cells.



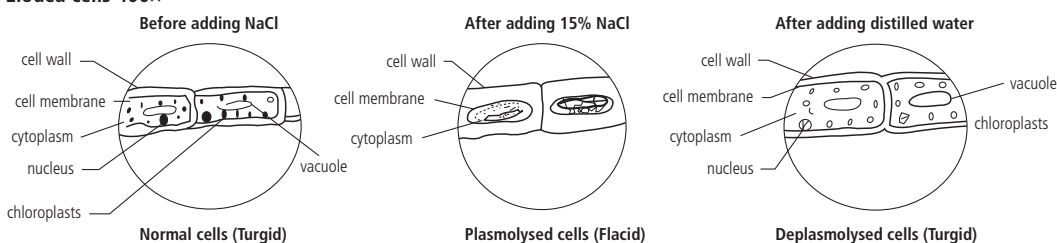
Plasmolysis

Results

Onion cells 400x



Elodea cells 400x



Analysis of results

1 If a plant cell is immersed in a hypertonic solution (salt solution), water will leave the cell. This process is called plasmolysis. When a plant cell is placed in hypotonic solution (distilled water) water re-enters the plasmolysed cell. This process is called deplasmolysis. **2** The cell shrinks during plasmolysis because water leaves the cell. **3** Both onion and elodea cells lost water when placed in 15% salt solution because the salt solution was hypertonic. **4** Both onion and elodea cells gained water when placed in distilled water because it was hypotonic. **5** Plant cells are isotonic to their environment so that they neither gain nor lose water. **6** Plant cells become turgid, when water enters the cells. **7** The cell wall prevents a plant cell from bursting. **8** Elodea is a fresh water plant because when it was placed in salt solution it loses water. **9** Grass dies because it loses water to its surroundings which have more salt. **10** Spraying helps to keep the vegetables fresh because water diffuses into the vegetable cells. **11** If the crew drink sea water they lose water and become dehydrated. **12** When sugar is sprinkled on strawberries, they lose water.

Conclusion

1 Distilled water is considered hypotonic because water diffuses into onion and elodea cells. Salt solution is considered hypertonic because water diffuses out of the cells.



Effect of surface area to volume ratio on cell size

Hypothesis

Smaller cells have more surface area to volume ratio than larger cells. Spherical and flattened shapes have more surface area to volume ratio. As the surface area to volume ratio increases, the cell will be able to obtain more nutrients.

5 Effect of surface area to volume ratio on cell size (cont'd)

Part A

Results

Table 1 – Before diffusion

Cube	Surface area	Volume	SA:V ratio	Reduced ratio
3 cm ³	$3 \times 3 \times 6 = 54$	$3 \times 3 \times 3 = 27$	54:27	2:1
2 cm ³	$2 \times 2 \times 6 = 24$	$2 \times 2 \times 2 = 8$	24:8	3:1
1 cm ³	$1 \times 1 \times 6 = 6$	$1 \times 1 \times 1 = 1$	6:1	6:1

Table 2 – After diffusion (Results could vary)

Cube	Dimensions of coloured cube (cm)	Volume of coloured cube (A)	Volume of block (B)	Volume of uncoloured portion (B – A)	Percentage of uncoloured portion $B - A \times 100 \div B$
3 cm ³	$l = 2, w = 2, h = 2$	$2 \times 2 \times 2 = 8$	$3 \times 3 \times 3 = 27$	$27 - 8 = 19$	$19 \times 100 \div 27 = 70\%$
2 cm ³	$l = 1, w = 1, h = 1$	$1 \times 1 \times 1 \times 1$	$2 \times 2 \times 2 = 8$	$8 - 1 = 7$	$7 \times 100 \div 8 = 87\%$
1 cm ³	$l = 0, w = 0, h = 0$	$0 \times 0 \times 0 = 0$	$1 \times 1 \times 1 = 1$	$1 - 0 = 1$	$1 \times 100 \div 1 = 100\%$

Analysis of results

1 The larger cube has greater surface area and volume, but its SA:V ratio is smaller. The smaller cube has greater SA:V ratio. **2** As the cube gets larger its area and volume increase but its SA:V ratio decreases. **3** As the cube gets larger its volume grows faster than its area. **4** When the 2 cm³ cube is cut in half, the surface area and volume of each half will decrease but the surface area to volume ratio increases as shown in the table below.

Cube dimensions	Surface area	Volume	SA:V ratio
2cm ³ cube	$2 \times 2 \times 6 = 24 \text{ cm}^2$	$2 \times 2 \times 2 = 8 \text{ cm}^3$	24:8 = 3:1
One half of 2 cm ³ cube	$(2 \times 2) \times 2 + (2 \times 1) \times 4 = 16 \text{ cm}^2$	$1 \times 2 \times 2 = 4 \text{ cm}^3$	16:4 = 4:1
Other half of 2 cm ³ cube	$(2 \times 2) \times 2 + (2 \times 1) \times 4 = 16 \text{ cm}^2$	$1 \times 2 \times 2 = 4 \text{ cm}^3$	16:4 = 4:1

Part B

Results

Table 3 – Before diffusion

Cell shape	Surface area	Volume	SA:V Ratio	Reduced ratio
Cube $l = 1 \text{ cm}$	$1 \times 1 \times 6 = 6$	$6 \times 1 \times 1 = 1$	6:1	6:1
Sphere $r = 0.5 \text{ cm}$	$4 \times 3.14 \times 0.5 \times 0.5 = 3.14$	$\frac{4}{3} \times 3.14 \times 0.5 \times 0.5 \times 0.5 = \frac{1.57}{3} = 0.52$	3:0.5	6:1
Cylinder $r = 0.5 \text{ cm}$, $h = 3 \text{ cm}$	$2 \times 3.14 \times 0.5 \times 3.5 = 10.99$	$3.14 \times 0.5 \times 0.5 \times 3 = 2.355$	11:2	5:1

Table 4: After diffusion

Results could vary

Cube	Dimensions of coloured cube (cm)	Volume of coloured cube (A)	Volume of block (B)	Volume of uncoloured portion (B – A)	Percentage of uncoloured portion $B - A \times 100 \div B$
cube, 1 cm ³	$l = 0, w = 0, h = 0$	$0 \times 0 \times 0 = 0$	$1 \times 1 \times 1 = 1$	$1 - 0 = 1$	$1 \times 100 \div 1 = 100\%$
sphere, $r = 0.5$	$r = 0$	$\frac{4}{3} \times 3.14 \times 0 = 0$	$\frac{4}{3} \times 3.14 \times 0.5 \times 0.5 \times 0.5 = 1.57$	$1.57 - 0 = 1.57$	$1.57 \times 100 \div 1.57 = 100\%$
cylinder, $r = 0.5$ $h = 3$, $\pi = 3.14$	$\pi r^2 h$ $r = 0.3, h = 2.0$	$3.14 \times 0.3 \times 0.3 \times 2 = 0.28$	$3.14 \times 0.5 \times 0.5 \times 3 = 2.355$	$2.355 - 0.28 = 2.075$	$2.075 \times 100 \div 2.355 = 88\%$

5 Effect of surface area to volume ratio on cell size (cont'd)

Analysis of results

1 The cylinder has the greater surface area and volume but has low SA:V ratio. 2 The sphere and the cube were more efficient shapes because their SA:V ratios were larger and the acid diffused faster. 3 The cell with smaller SA:V ratio would be in greatest danger of starving because it takes longer for nutrients to diffuse. 4 Smaller animals like mice have greater SA:V ratio and lose more heat. To maintain their body heat, they need to eat more to maintain a high metabolic rate. 5 Elephants tend to maintain their temperatures and get cooler or warmer slowly because they have a lower SA:V ratio. Therefore, their metabolic rate is slower and they eat less.

Conclusion

1 Unicellular organisms are usually spherical and flattened in shape so that they can obtain nutrients and gases efficiently. 2 A growing cell splits to increase its surface area to volume ratio. 3 As the size of the cube increases, the time taken for a substance to diffuse to the centre of the cube also increases. 4 As the size of the cell increases, its SA:V ratio decreases and it will take longer for the cell to get its nutrients and get rid of wastes.



6 Factors affecting cell membrane permeability

Part A

Hypothesis

Concentrated detergent solutions and alcohol affect the structure of the beetroot cell membrane and more pigment will leak out.

Results

Table 1

(Results could vary)

Time observed	Distilled water	Colour of the solutions									
		pH					Alcohol			Detergent	
		pH 2	pH 4	pH 6	pH 8	pH 10	1%	25%	50%	1%	5%
At the start	1	5	4	1	4	5	2	4	5	2	5
After 1 hr	1	5	4	1	4	5	2	4	5	2	5
After 24 hrs	1	5	4	1	4	5	2	4	5	2	5

Analysis of results

1 The dish with distilled water was used as the control. No changes occurred in the beetroot cells and the colour of the water remained clear. 2 Changes in the pH of the solutions, concentrations of the detergent and alcohol solutions were used as independent variables. 3 The leakage of the pigment was the dependent variable. It was measured by the colour intensity of the solution. The assumption is that the darker the colour of the solution, the greater the damage to the beetroot cell membrane. 4 The temperature, length and mass of the beetroot cylinders were controlled. 5 pH 6–7 was the ideal range for the beetroot cell membrane. At a pH outside this range, the protein molecules in the membrane were denatured and the pigment leaked out. 6 The pigment leaked out the most at pH 2, 4 and 10. 7 50% alcohol affected the membrane the most. 8 25% detergent affected the membrane the most. 9 The membrane was immediately affected by 25% detergent, 50% alcohol and pH 2 and pH 10 solutions. 10 The colour leaking out was from the vacuole of the beetroot cell. 11 After 24 hours, the beetroot cylinders appeared fresh and turgid in distilled water because water diffused into the cells.

Part B

Hypothesis

High temperatures affect the structure of the beetroot cell membrane, causing more pigment to leak out.

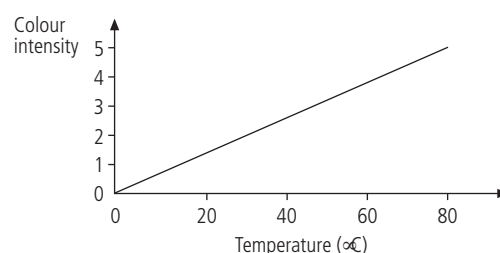
Results

Table 2 (Results could vary)

Temperature (°C)	Colour intensity of solutions (arbitrary units)
Control: room temp.	0
0	1
20	2
40	3
60	4
80	5

Analysis of results

1 The effect of temperature on colour intensity



6 Factors affecting cell membrane permeability (cont'd)

2 The test tube containing distilled water at room temperature was used as the control. **3** It is possible to tell the degree to which the membrane has been damaged by looking at the amount of pigment leakage. **4** The membrane did not suffer any damage at 0°C to 20°C temperatures. **5** Most of the colour leaked out between the temperatures 40°C and 80°C. **6** At higher temperatures, the cell membrane structure was disrupted because the protein molecules were denatured by the heat. **7** This reaction is not reversible because once the protein molecules are denatured, their shape cannot be changed. **8** Sources of errors and suggestions for improvement. **a** The beetroot cylinders need to be thin and the masses must be identical. **b** Maintaining the temperature of the water baths at regular intervals is essential. **c** Giving arbitrary units for colour intensity is subjective and not quantitative. Using a colorimeter will give quantitative results.

Conclusion

1 Changes that occurred to the lipid layer and protein molecules in the cell membranes when beetroot pieces were placed in the following solutions. **a** Distilled water: the cell membrane was not affected and the pigment did not leak out of the cells. **b** Different pH solutions: the protein molecules were denatured and the pigment leaked out of the cells. **c** Different strengths of alcohol: stronger alcohol solutions affected the fatty acids in the cell membrane and the pigment leaked out of the cells. **d** Different strengths of detergent: stronger detergent solutions affected the fatty acids in the cell membrane and the pigment leaked out of the cells. **2** At higher temperatures the protein molecules were denatured and more pigment leaked out of the beetroot cells. **3** The results from part A and part B supported the hypotheses.



7 Factors affecting catalase activity

Part A

Hypothesis

The enzyme catalase contained in plant and animal cells, under optimum conditions, effectively speeds the decomposition of hydrogen peroxide.

Results

Table 1 – Catalase reactions with hydrogen peroxide

Test tube	Materials used	Rate of reaction
A (control)	sand + H ₂ O ₂	0
B	whole liver + H ₂ O ₂	4
C	ground liver + H ₂ O ₂	5
D	boiled liver + H ₂ O ₂	0
E	potato + H ₂ O ₂	4
F	chicken + H ₂ O ₂	4
G	apple + H ₂ O ₂	4

Analysis of results

1 Test tube A with sand + H₂O₂ was used as the control. **2** The ground liver produced more bubbles than the whole liver because more catalase was released to react with the hydrogen peroxide. **3** Boiling the liver denatured the enzyme catalase and it did not react with hydrogen peroxide. **4** Catalase decomposed the hydrogen peroxide into water and oxygen gas. $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ **5** All the cells tested contained catalase. When cells from chicken, liver, apple and potato were added to hydrogen peroxide, the enzyme catalase decomposed the hydrogen peroxide.

Part B

Hypothesis **1** At temperatures above the optimum temperature, the enzyme denatures and the enzyme activity stops. **2** At pH above or below the optimum pH, the enzyme denatures and the enzyme activity stops. **3** As the concentration of the substrate increases, the rate of enzyme activity increases up to a certain extent and then levels off. **4** As the concentration of the enzyme increases, the rate of enzyme activity increases if the amount of substrate is unlimited.

Results

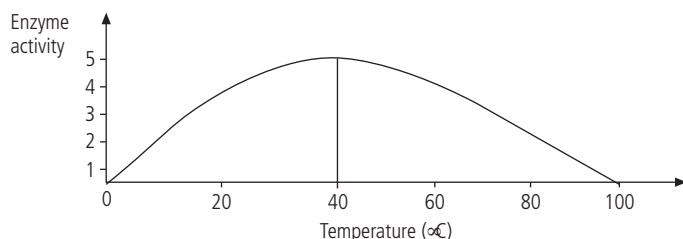
Tables 1, 2, 3, 4 Answers could vary.

Temperature (°C)	Enzyme activity	pH	Enzyme activity	Amount of substrate H ₂ O ₂	Enzyme activity	Amount of liver catalase (g)	Enzyme activity
0	1	1	0	5 cc	1	0.1	1
20	3	3	0	10 cc	3	0.2	2
40	5	5	4	15 cc	4	0.3	3
60	3	7	5	20 cc	5	0.4	4
80	0	9	4	25 cc	5	0.5	5
100	0	11	0	30 cc	5	0.6	5
		13	0	35 cc	5		

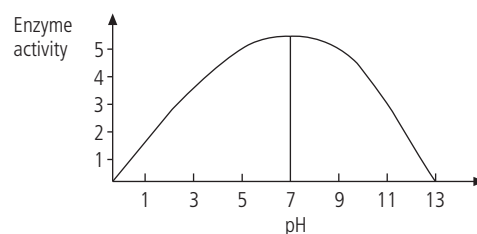
7 Factors affecting catalase activity (cont'd)

Graphing

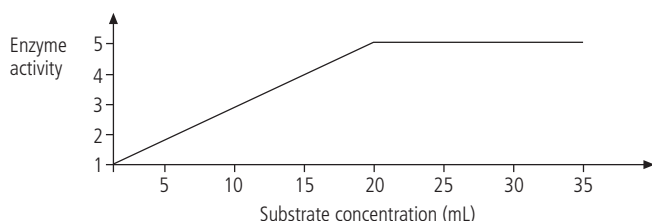
Effect of temperature on catalase activity
Optimum temperature 40°C



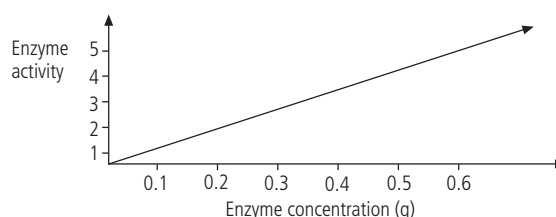
Effect of pH on catalase activity
Optimum pH: 7



Effect of substrate concentration on catalase activity



Effect of enzyme concentration on catalase activity



Analysis of results

Effect of temperature

1 The enzyme reacted most vigorously at 40°C. The enzyme reacted least vigorously between 60°C and 100°C. **2** 40°C is the optimum temperature for catalase. **3** As the temperature increased from 0°C to 20°C, the rate of reaction increased. The rate of reaction decreased and stopped as the temperature increased from 60°C to 100°C. **4** At higher temperatures, the enzyme is denatured and it cannot react with the substrate.

Effect of pH

5 The enzyme reacted most vigorously at pH 7. The enzyme reacted least vigorously at pH 1, 3, 11 and 13. **6** pH 7 is the optimum pH for catalase. **7** The reaction slowed down and stopped at pH below 7 and at pH above 7. **8** The enzyme was denatured at pH levels above and below 7 and did not react with the substrate.

Effect of substrate concentration

9 As the amount of H₂O₂ increased, the rate of reaction increased to a certain extent and then levelled off. **10** The rate of enzyme reaction levelled off because there were not enough enzyme molecules to react with the substrate molecules.

Effect of enzyme concentration

11 As the concentration of enzyme increased, the rate of reaction increased. **12** The enzyme can be reused because it is not affected by the reaction.

Conclusion

1 The rate of reaction is much faster with ground liver than with whole liver. When the liver was boiled, the enzyme was denatured and did not react with the hydrogen peroxide. **2** Catalase decomposes hydrogen peroxide to water and oxygen: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ **3** The optimum temperature for catalase is 40°C and optimum pH is pH 7. **4** As the substrate increases, the rate of reaction increases to a certain extent and then levels off. **5** As the amount of enzyme increases, the rate of reaction increases. **6** The experiment could be improved by **a** having a control and varying only the one condition that is to be tested. **b** keeping all the other variables constant. **c** repeating the experiment and collating class results. **d** using fresh liver and freshly-made hydrogen peroxide. **e** using a meter to measure the oxygen released.



8 Factors affecting the rate of photosynthesis

Part A

Hypothesis

If the light intensity increases, the rate of photosynthesis will increase at a proportional rate until a certain level is reached. Eventually, a level will be reached where an increase in light intensity will have no further effect on the rate of photosynthesis due to other limiting factors such as temperature and carbon dioxide.

Results

Lamp distance (cm)	Oxygen evolved – length (cms)	
	Light intensity in lux	Average
5	1015	very high
10	945	↓
15	590	
20	310	high
25	208	medium

Lamp distance (cm)	Oxygen evolved – length (cms)	
	Light intensity in lux	Average
30	149	low
35	110	↓
40	80	
45	55	very low

8 Factors affecting catalase activity (cont'd)

Analysis of results

1 Oxygen gas bubbles were released from the elodea plant when it was placed in light. This demonstrates that plants give off oxygen during photosynthesis. **2** As the light intensity increased, the rate of photosynthesis increased to a certain extent and then other limiting factors such as carbon dioxide and temperature slowed down the rate of photosynthesis. The relationship between light intensity and the rate of photosynthesis was non-linear and increased at an exponential rate. **3** By adding sodium bicarbonate to the water, the plant was supplied with plenty of CO_2 . **4** Suggestions for improvement **a** The distances between the light sources and elodea plant need to be measured to a very high degree of accuracy. The distance must be measured from the light filament to the centre of the elodea plant. **b** It would be better to take more readings at 5, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40 and 45 cms distances to get accurate results. **c** Errors in reading the volume of oxygen in the syringe. It would be necessary to do the readings over a longer period of time to reduce the percentage of errors. **d** Background light in the area could have affected the results. To eliminate this problem completely, it would be necessary to perform the experiment in a completely dark room. **e** Heat generated by the lamp could have increased the temperature of water in the beaker. To avoid this, the temperature of water should be checked regularly and kept constant. **f** Since sodium bicarbonate was added to the water, carbon dioxide was not the limiting factor. But if the experiment was conducted over a longer period of time its concentration must also be kept constant. **g** Increase the time of observation so that more oxygen can be collected.

Part B

Hypothesis

As the concentration of carbon dioxide increases, the rate of photosynthesis also increases and more starch is produced.

Results

Table 2 – Effect of carbon dioxide on starch production

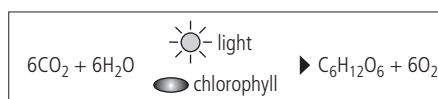
Plants	Colour of the leaf after adding potassium iodide	Amount of starch Present
Control plant A: no CO_2	brown	No starch present
Experimental plant B: with CO_2	dark blue-black	More starch present

Analysis of results

1 As sodium hydroxide absorbed CO_2 from the air in the bell jar, plant A was deprived of CO_2 and this reduced the rate of photosynthesis. As a result, no starch was produced by plant A. **2** As sodium bicarbonate released more CO_2 for plant B to use, its rate of photosynthesis increased. As a result, it produced more starch. **3** Suggestions for improvement **a** Even though bell jars were used, the amount of CO_2 around the plants may not have been controlled completely because the seal around the bell jar may not have been airtight. **b** The light intensity may not have been controlled properly if the room was not dark enough. It is better to do this experiment in a darkened room. **c** Temperature variations over the 24 hours could have affected the rate of photosynthesis. Temperature should be maintained as a constant by doing this experiment in a room with a thermostat where the temperature is set to 25°C .

Conclusion

1 Photosynthesis is the process by which green plants use sunlight to produce glucose and oxygen from simple compounds such as water and carbon dioxide.



2 As the light intensity increased, the rate of photosynthesis also increased until no further increase occurred because other limiting factors such as temperature and CO_2 affected the rate of photosynthesis. As the carbon dioxide concentration increased, the rate of photosynthesis also increased and more starch was produced. These results supported my hypothesis.



Aerobic and anaerobic respiration

Part A

Hypothesis

Germinating pea seeds will release heat and CO_2 during aerobic respiration and dry seeds will not.

Results

Table 1 In jar A (containing dry seeds), the temperature and the lime water colour did not change. In Jar B, the temperature increased and the lime water turned milky as CO_2 was produced by germinating seeds.

9 Aerobic and anaerobic respiration (cont'd)

1 The jars were packed in sawdust to insulate them so that no heat is gained or lost. **2** Jar A with dry pea seeds was used as the control. **3** The presence of germinating pea seeds was the independent variable. Temperature increase and release of CO₂ were the dependent variables. **4** The temperature rise in jar B indicated that germinating seeds released heat energy. **5** CO₂ was produced during the aerobic respiration of germinating pea seeds. **6** The lime water turned milky when CO₂ from jar B bubbled through the lime water but no change occurred in lime water when gas from jar A passed through the lime water.

Part B

Hypothesis

Yeast will release heat and produce CO₂ and ethanol during anaerobic respiration.

Results

Table 2 – In flask A (the control flask) no changes were observed. In flask B the temperature increased. The lime water turned milky and the glucose solution turned turbid and smelt of alcohol.

Analysis of results

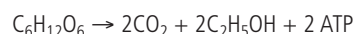
1 The conical flask A (containing glucose without yeast) was used as the control so that the results from the experimental set up could be compared. **2** The presence of yeast was used as the independent variable and the production of heat, CO₂ and ethanol were the dependent variables. **3** The increase in the temperature and the production of CO₂ indicates that fermentation is occurring. **4** After 60 minutes the temperature increased and the lime water turned milky. **5** As the lime water turned milky, it could be inferred that CO₂ was released. **6** In the control flask A no changes occurred in the glucose. In the experimental flask B, the glucose became turbid and smelt of alcohol. **7** Suggestions for improvement **a** The experimental set up must be airtight so that no heat is lost and CO₂ does not escape. **b** Oxygen from water, glucose solution and flasks must be removed to prevent aerobic respiration. **c** Maintain the temperature of the water bath at 40°C throughout the experiment. Temperature affects the respiratory enzymes in the yeast. Thermos flasks can be used instead of conical flasks. **d** The amount of yeast and glucose must be kept constant because they also have an effect on the results.

Conclusion

Aerobic respiration took place in germinating pea seeds in the presence of oxygen. Glucose was completely broken down to CO₂, H₂O and more heat energy was released.



Anaerobic respiration took place in yeast cells in the absence of oxygen. During fermentation, glucose was partially broken and ethanol, CO₂ and small amount of heat were produced.



Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing the **highest quality learning outcomes with instant usability**.

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Practical Experiments in Biology: Book 1

This book is part of a four-book series that provides self-contained practical Biology experiments for senior students. Each experiment contains step-by-step instructions, diagrams and safety issues. They contain multiple parts that can be used as stand-alone experiments, or an extended experiment. There are also recommended websites and suggestions for teachers relating to assumed knowledge and experimental technique. Suggested answers and expected results have been provided where appropriate.

Topics include

Introduction to light microscope
Observing cells
Diffusion and osmosis
Plasmolysis
Effect of surface area to volume ratio on cell size

Factors affecting cell membrane permeability
Factors affecting catalase activity
Factors affecting the rate of photosynthesis
Aerobic and anaerobic respiration

Associated titles in the Instant Lessons series

- Practical Experiments in Biology (Books 2, 3 & 4)
- Assignments in Biology (Books 1–3)
- Interpreting Data in Biology (Books 1 & 2)
- Assignments in Genetics (Books 1 & 2)
- Instant Lessons in Biology (Books 1–3)

learning
ESSENTIALS
Bringing you the best!

Instant Lessons

An imprint of Learning Essentials
A.B.N. 56 093 980 726
108 Main Rd
Clayton South VIC 3169
Ph: +61 3 9548 0193
www.learningessentials.com.au

ISBN 978-1-92561-010-9



9 781925 610109